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(54) **WEARABLE MEDICAL SYSTEM (WMS)
SYSTEM DETECTING BRADYCARDIA, AND
CONFIRMING THE DETECTION BEFORE
PACING TO TREAT THE DETECTED
BRADYCARDIA**

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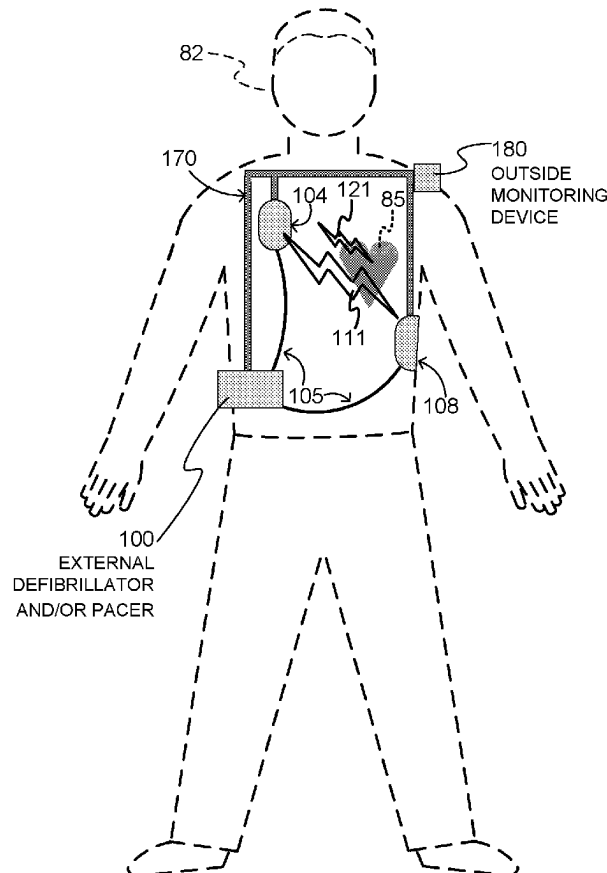
(57) **ABSTRACT**

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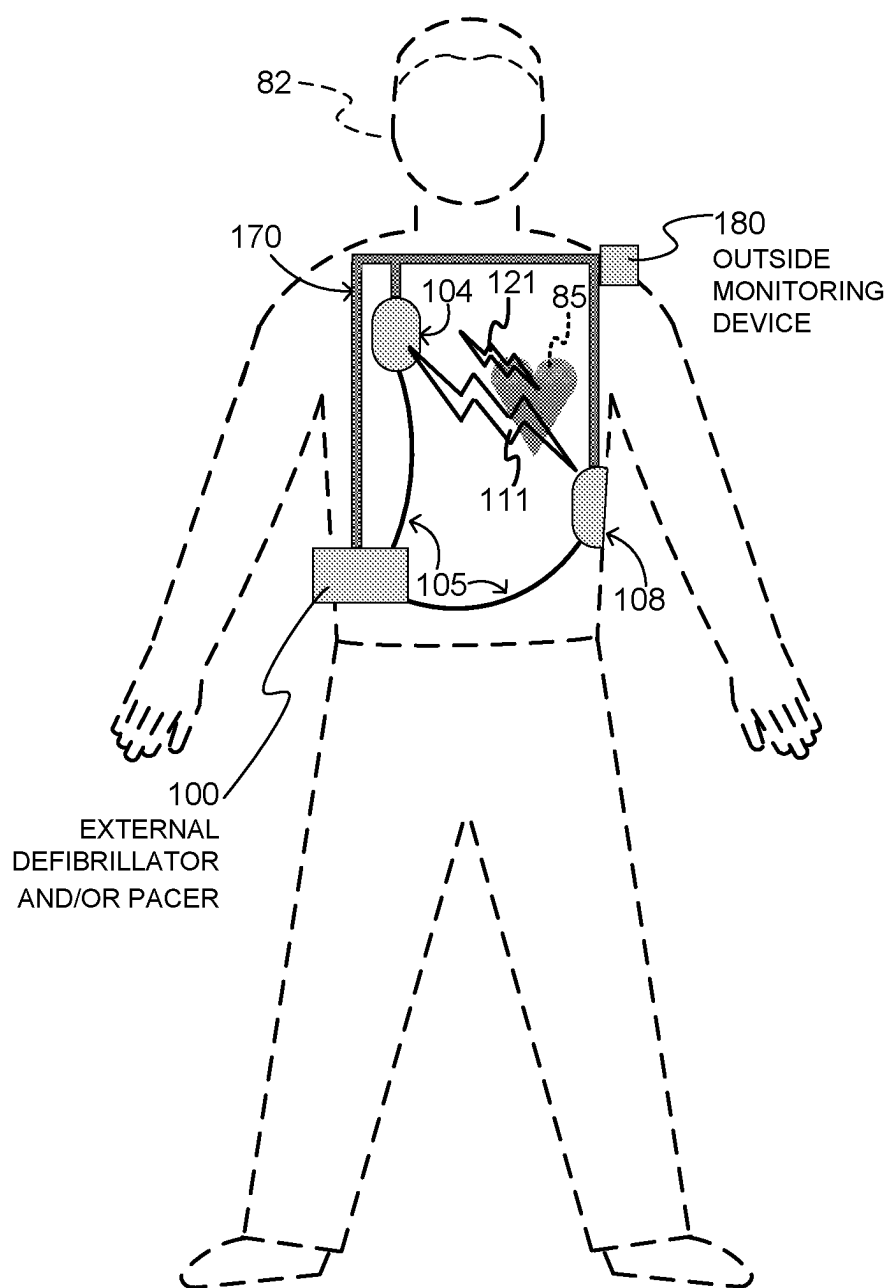
A Wearable Medical System (WMS) includes one or more pacing capabilities. The WMS may detect when the patient's heart rhythm starts to deteriorate, but not necessarily in a way that requires defibrillation. In particular, the WMS may detect bradycardia of one or more types, and then confirm the detection before pacing to treat the detected bradycardia.

Related U.S. Application Data

(63) Continuation of application No. 17/561,607, filed on Dec. 23, 2021, now Pat. No. 12,329,973.

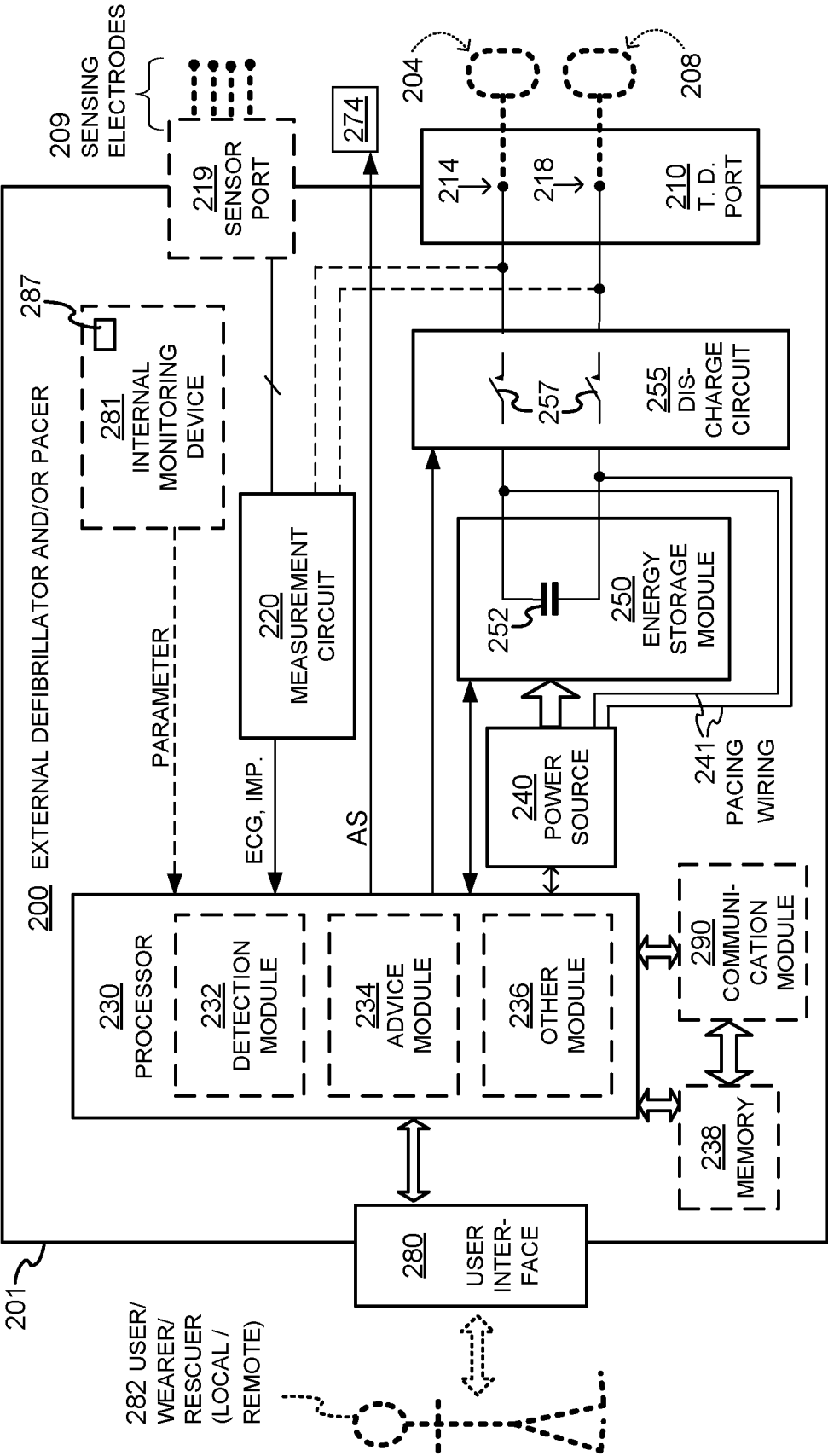


**SAMPLE COMPONENTS OF WEARABLE
MEDICAL SYSTEM (WMS)**



SAMPLE COMPONENTS OF WEARABLE
MEDICAL SYSTEM (WMS)

FIG. 1



SAMPLE COMPONENTS OF EXTERNAL DEFIBRILLATOR
AND/OR PACER

FIG. 2

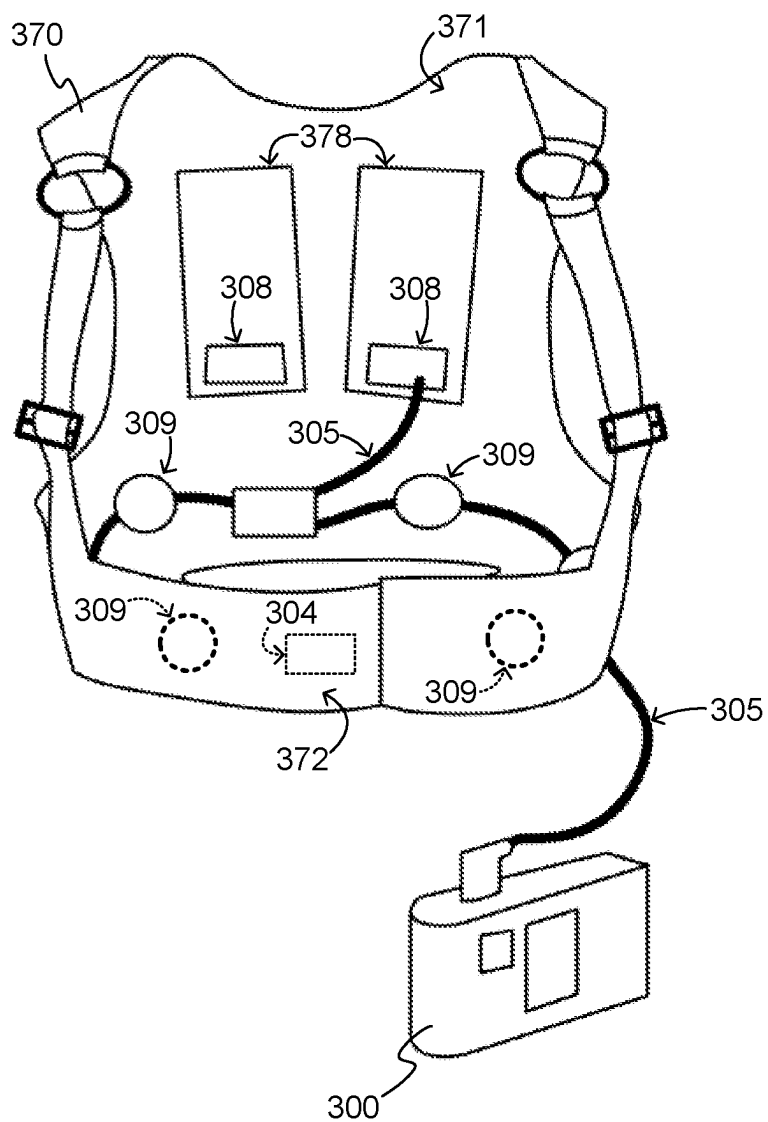


FIG. 3

COMPONENTS OF
SAMPLE WMS

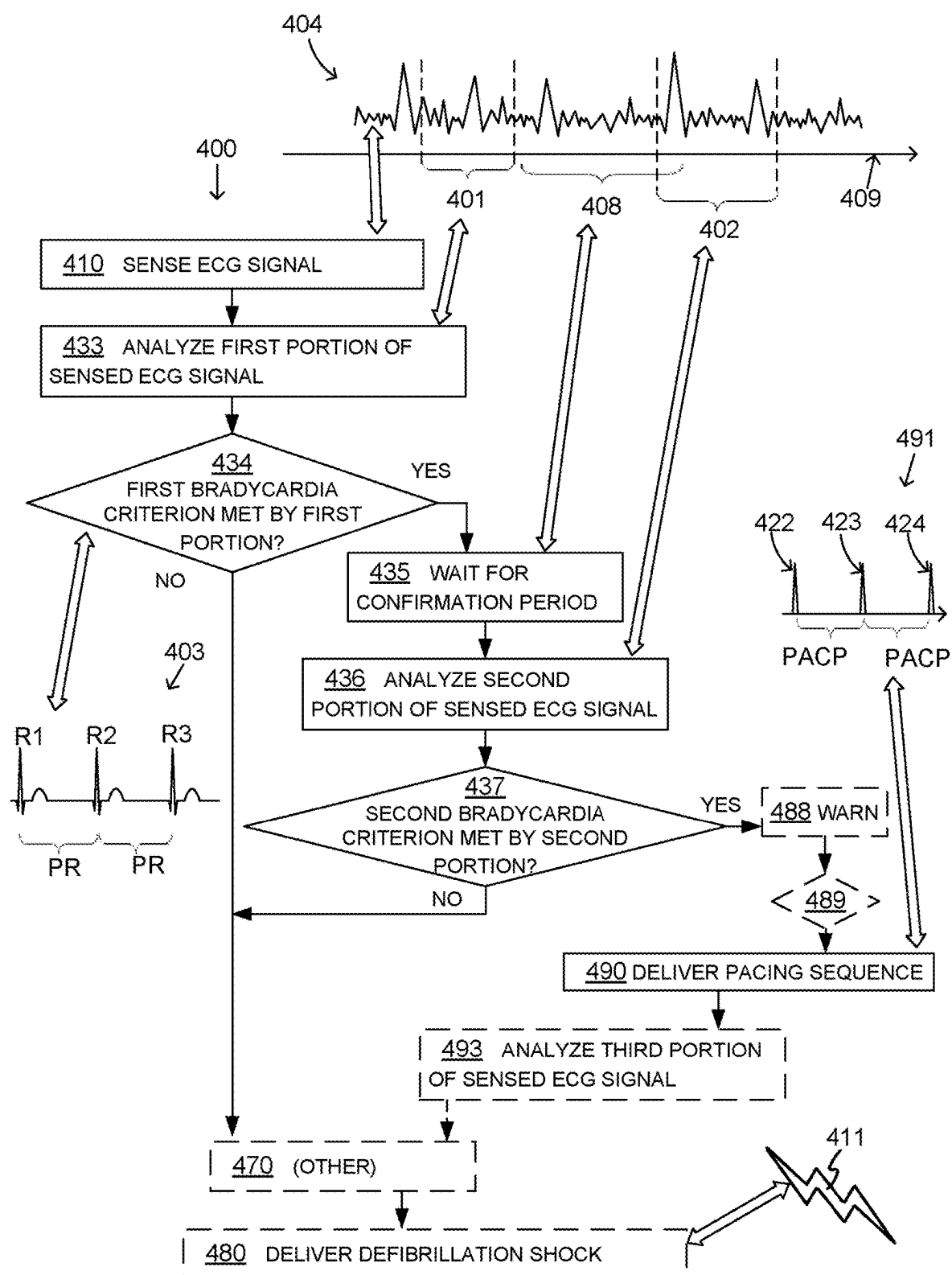


FIG. 4

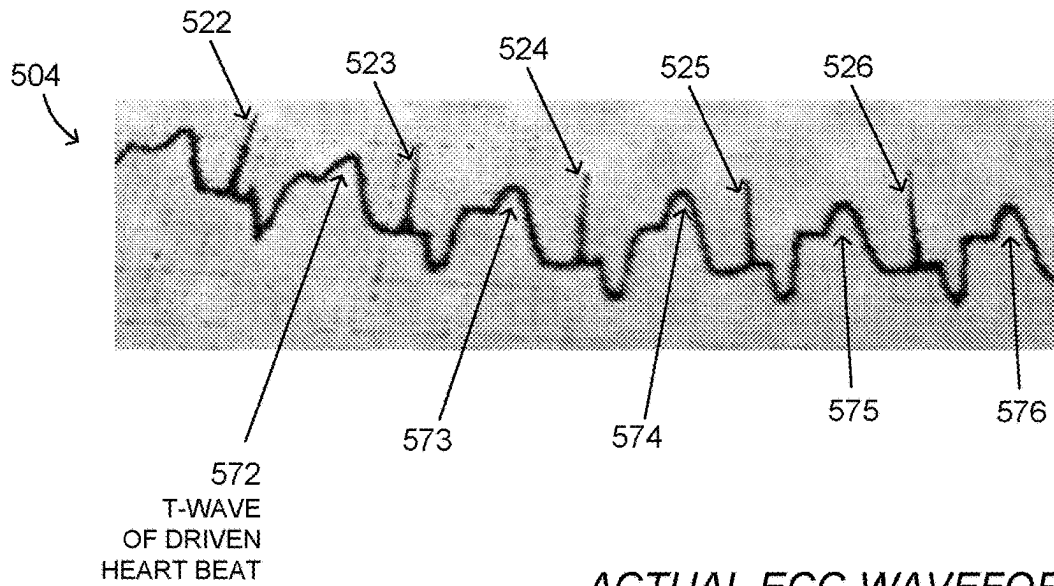


FIG. 5

ACTUAL ECG WAVEFORM
WITH PACING AND NO NOISE

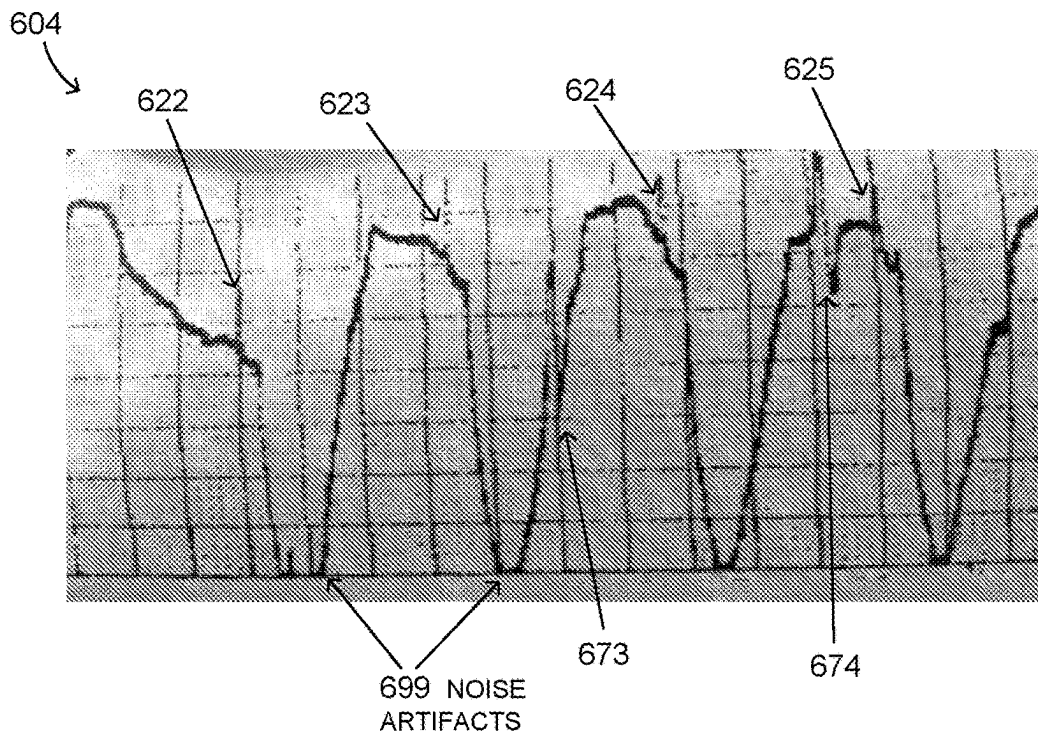


FIG. 6

ACTUAL ECG WAVEFORM
WITH PACING AND NOISE

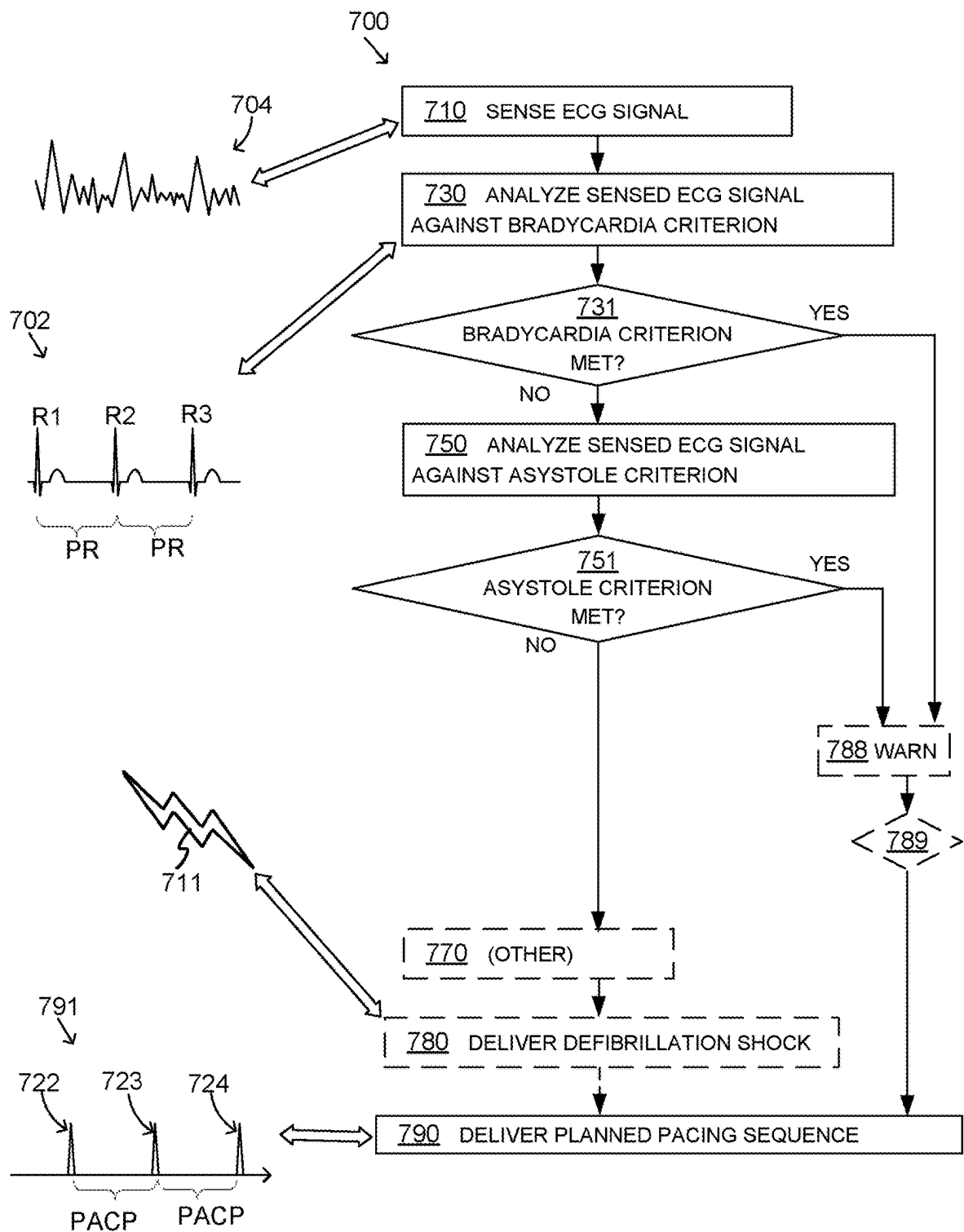


FIG. 7

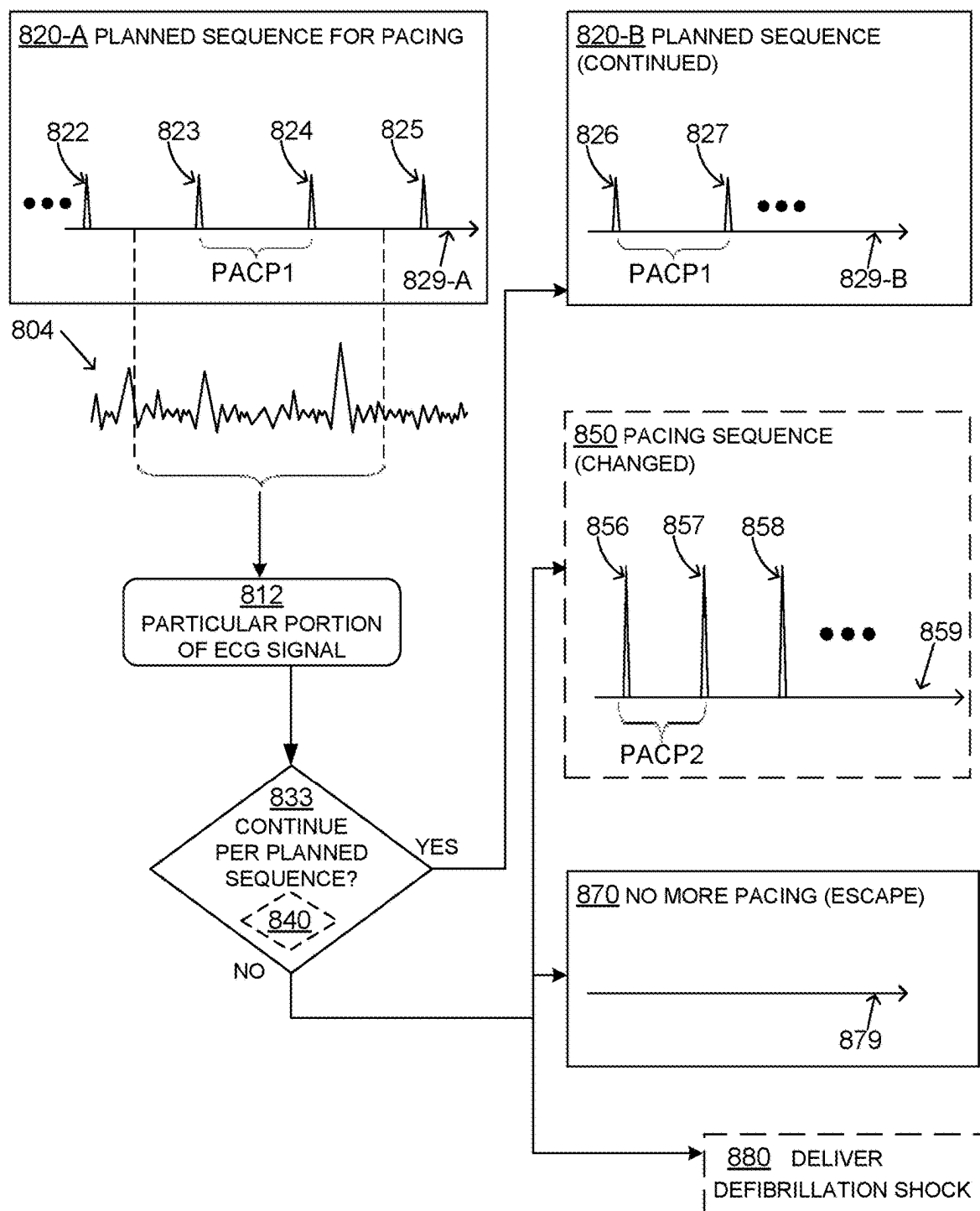
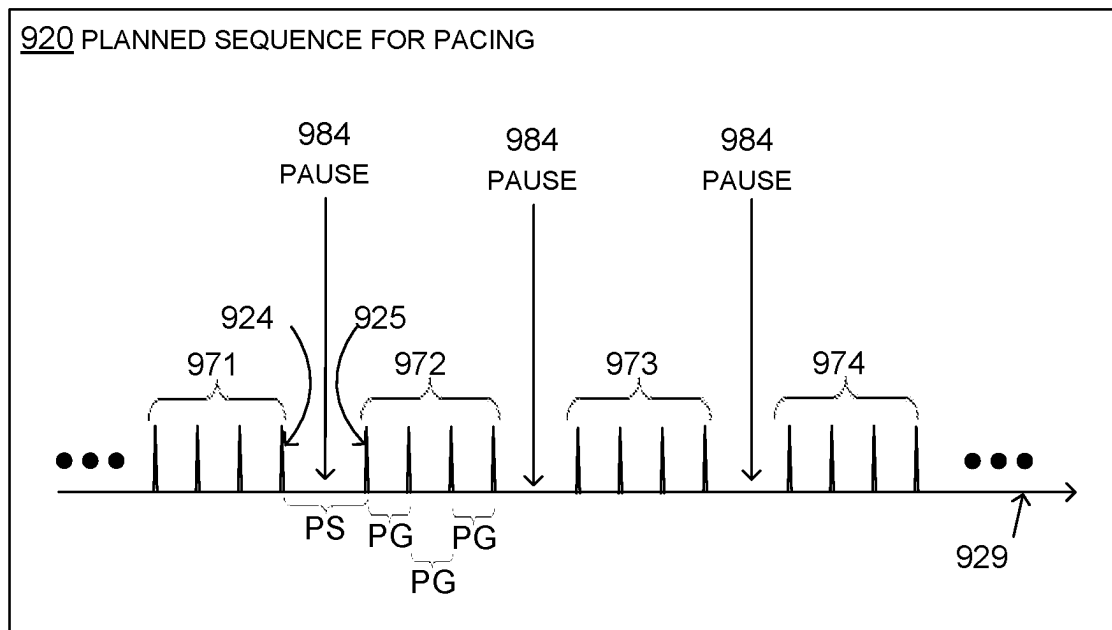


FIG. 8



PACING PULSES IN PERIODIC GROUPS
WITH PAUSES BETWEEN THE GROUPS

FIG. 9

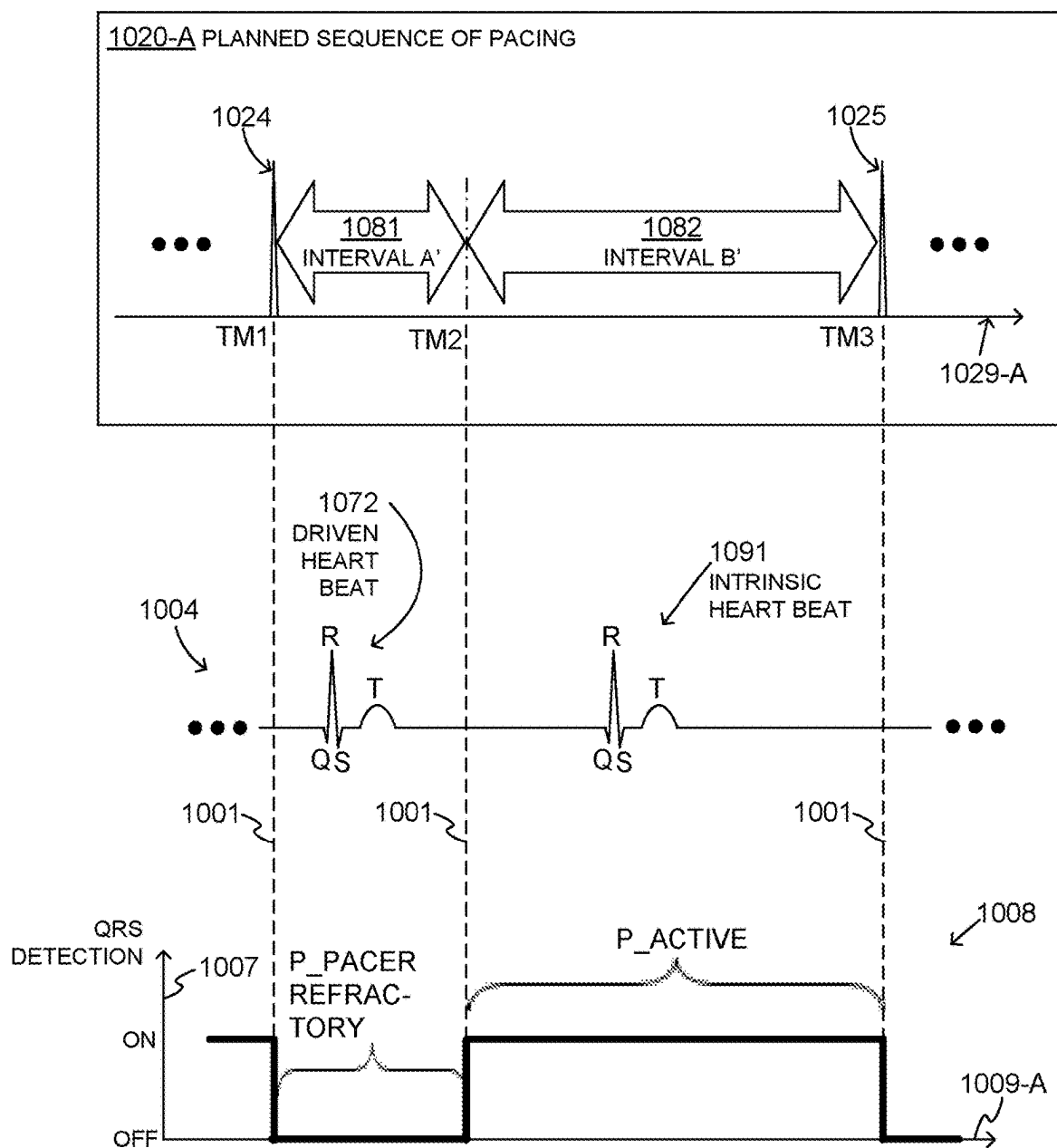


FIG. 10

DETECTING QRS COMPLEXES
IN ECG SIGNAL PORTIONS
DURING THE INTERVALS
(IDEALIZED WAVEFORMS)

1100

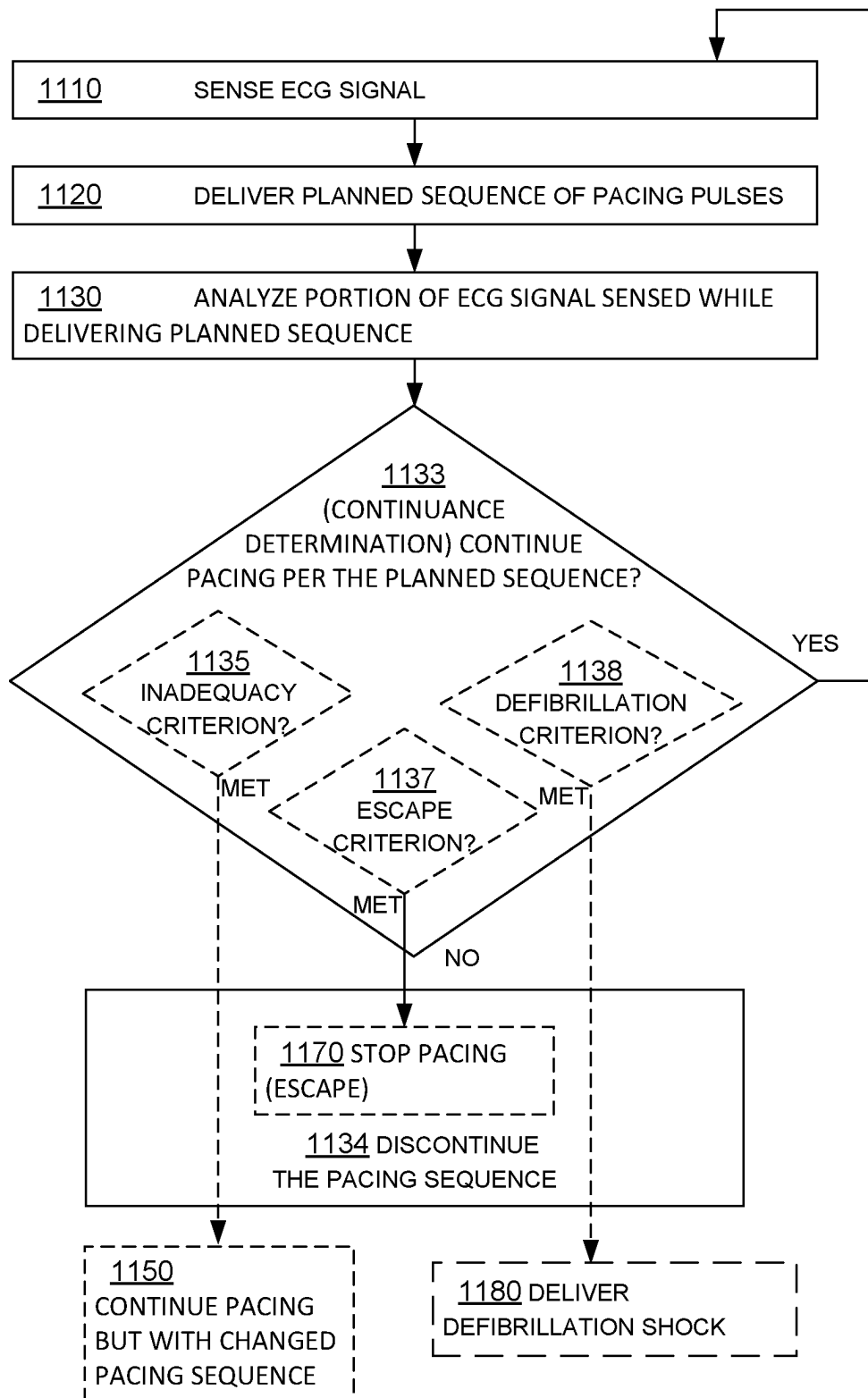


FIG. 11

METHODS

**WEARABLE MEDICAL SYSTEM (WMS)
SYSTEM DETECTING BRADYCARDIA, AND
CONFIRMING THE DETECTION BEFORE
PACING TO TREAT THE DETECTED
BRADYCARDIA**

**CROSS REFERENCE TO RELATED PATENT
APPLICATIONS**

[0001] This patent application is a continuation patent application of U.S. patent application Ser. No. 17/561,607, filed on Dec. 23, 2021, entitled “RHYTHM SENSING DURING EXTERNAL PACING”, which claims priority from U.S. Provisional Patent Application Ser. No. 63/129,887, filed on Dec. 23, 2020; the disclosures of which are hereby incorporated by reference for all purposes.

BACKGROUND

[0002] When people suffer from some types of heart arrhythmias, the result may be that blood flow to various parts of the body is reduced. Some arrhythmias may even result in a Sudden Cardiac Arrest (SCA). SCA can lead to death very quickly, e.g. within 10 minutes, unless treated in the interim. Some observers may have thought that SCA is the same as a heart attack, but it is not.

[0003] Some people have an increased risk of SCA. Such people include patients who have had a heart attack, or a prior SCA episode. A frequent recommendation for these people is to receive an Implantable Cardioverter Defibrillator (ICD). The ICD is surgically implanted in the chest, and continuously monitors the patient's electrocardiogram (ECG). If certain types of heart arrhythmias are detected, then the ICD delivers an electric shock through the heart.

[0004] As a further precaution, people who have been identified to have an increased risk of an SCA are sometimes given a Wearable Cardioverter Defibrillator (WCD) system, to wear until the time that their ICD is implanted. Early versions of such systems were called wearable cardiac defibrillator systems. A WCD system typically includes a harness, vest, belt, or other garment that the patient is to wear. The WCD system further includes electronic components, such as a defibrillator and electrodes, coupled to the harness, vest, or other garment. When the patient wears the WCD system, the electrodes may make good electrical contact with the patient's skin, and therefore can help sense the patient's ECG. If a shockable heart arrhythmia is detected from the ECG, then the defibrillator delivers an appropriate electric shock through the patient's body, and thus through the heart. This may restart the patient's heart, and thus save their life.

[0005] WCDs have far harder challenges to overcome, than regular monitor-defibrillators that are carried by trained professionals, or AEDs (Automated External Defibrillators) that may be used by a bystander. In both the scenarios of a monitor-defibrillator and an AED, the patient is typically motionless. A WCD does not necessarily have an operator and, in such situations, it must operate autonomously-make detections, deliver treatments and so on. For detection, a WCD may have far worse challenges due to electrical noise, since the patient may be conscious before SCA, moving, which generates electrical noise, and/or there is no operator to ensure good electrical contacts for the electrodes to suppress the electrical noise.

[0006] All subject matter discussed in this Background section of this document, including recognition of challenges, is not necessarily prior art, and may not be presumed to be prior art simply because it is presented in this Background section. Plus, any reference to any prior art in this description is not, and should not be taken as, an acknowledgement or any form of suggestion that such prior art forms parts of the common general knowledge in any art in any country. Along these lines, any recognition of problems in the prior art discussed in this Background section or associated with such subject matter should not be treated as prior art, unless expressly stated to be prior art. Rather, the discussion of any subject matter in this Background section should be treated as part of the approach taken towards the particular problem by the inventors. This approach in and of itself may also be inventive.

BRIEF SUMMARY

[0007] The present description gives instances of Wearable Medical Systems (WMSs), storage media that may store programs, and methods, the use of which may help overcome problems and limitations of the prior art.

[0008] In embodiments, a WMS includes one or more pacing capabilities. The WMS may detect when the patient's heart rhythm starts to deteriorate, but not necessarily in a way that requires defibrillation. In particular, the WMS may detect bradycardia of one or more types, and then confirm the detection before pacing to treat the detected bradycardia.

[0009] In embodiments, a WMS includes one or more pacing capabilities. The WMS may detect when the patient's heart rhythm starts to deteriorate, but not necessarily in a way that requires defibrillation. In particular, the WMS applies the same pacing sequence to treat bradycardia and asystole. Optionally, it also applies this pacing sequence after a defibrillation shock, to treat post-shock bradycardia or perhaps repeating events of asystole.

[0010] In embodiments, a WMS includes one or more pacing capabilities. The WMS may detect when the patient's heart rhythm starts to deteriorate, but not necessarily in a way that requires defibrillation. In particular, the WMS may pace the patient and concurrently monitor for arrhythmias. Upon detecting an arrhythmia, the WMS may continue, adjust, or discontinue the pacing.

[0011] As such, it will be appreciated that results of embodiments are larger than the sum of their individual parts, and have substantial utility.

[0012] These and other features and advantages of the claimed invention will become more readily apparent in view of the embodiments described and illustrated in this specification, namely in this written specification and the associated drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is a diagram of sample components of a wearable medical system (WMS), which is made according to embodiments.

[0014] FIG. 2 is a diagram showing sample components of an external defibrillator and/or pacer, such as the one of the WMS of FIG. 1, and which is made according to embodiments.

[0015] FIG. 3 is a diagram of sample embodiments of components of a WMS.

[0016] FIG. 4 shows a flowchart for illustrating sample methods according to embodiments, and it further augments the flowchart with small diagrams and/or icons of elements that can be related to individual operations of the flowchart.

[0017] FIG. 5 is a diagram of a waveform of a sample actual ECG signal, sensed while the patient is being paced according to embodiments, and with no appreciable electrical noise present.

[0018] FIG. 6 is a diagram of a waveform of a sample actual ECG signal, sensed while the patient is being paced according to embodiments, and with electrical noise present.

[0019] FIG. 7 shows a flowchart for illustrating sample methods according to embodiments, and it further augments the flowchart with small diagrams and/or icons of elements that can be related to individual operations of the flowchart.

[0020] FIG. 8 shows a time diagram of a sample planned sequence for pacing and in which the ECG signal is detected concurrently with the pacing, and it further shows various possibilities of how the pacing may be continued per the planned sequence, or the sequence may be adjusted or stopped responsive to analyzing the detected ECG signal, according to embodiments.

[0021] FIG. 9 is a time diagram of a sample planned pacing sequence that has groups of pacing pulses that are delivered substantially periodically, and in particular are delivered in groups of pulses that are separated by longer pauses, according to embodiments.

[0022] FIG. 10 shows elements of a technique for detecting intrinsic heartbeats but excluding heartbeats driven by the pacing according to embodiments.

[0023] FIG. 11 is a flowchart for illustrating sample methods according to embodiments.

DETAILED DESCRIPTION

[0024] As has been mentioned, the present description is about Wearable Medical Systems (WMSs), storage media that may store programs, and methods. Such systems may include a pacemaker and/or a defibrillator. If they include both, they may also be called Wearable Cardioverter Pacemaker Defibrillator (WPCD) systems. If they include only a defibrillator, they may also be called Wearable Cardioverter Defibrillator (WCD) systems. If they include only a pacemaker, which is also known as a pacemaker, they may also be called Wearable Pacer (WP) systems. Embodiments are now described in more detail.

[0025] A wearable medical system (WMS) according to embodiments may protect an ambulatory patient by electrically restarting their heart if needed. Such a WMS may have a number of components. These components can be provided separately as modules that can be interconnected, or can be combined with other components, and so on.

[0026] FIG. 1 depicts a patient 82. The patient 82 may also be referred to as the person 82 and/or wearer 82, since the patient 82 is wearing components of the WMS. The patient 82 is ambulatory, which means that, while wearing the wearable portion of the WMS, the patient 82 can walk around and is not necessarily bed-ridden. While the patient 82 may be considered to be also a “user” of the WMS, this definition is not exclusive to the patient 82. For instance, a user of the wearable medical system (WMS) may also be a clinician such as a doctor, nurse, emergency medical technician (EMT) or other similarly tasked individual or group of individuals. In some cases, a user may even be a

bystander. The particular context of these and other related terms within this description should be interpreted accordingly.

[0027] In some embodiments a WMS is configured to also defibrillate the patient who is wearing the designated parts the WMS. Defibrillating can be by the WMS delivering an electrical charge to the patient's body in the form of an electric shock. This electric shock is also known as a defibrillation shock, and its energy can be more than 100 Joule (J), such as 200 J, 300 J, 360 J and so on. The electric shock can be delivered in one or more pulses.

[0028] In particular, FIG. 1 also depicts components of a WMS made according to embodiments. One such component is a support structure 170 that is wearable by the ambulatory patient 82. Accordingly, the support structure 170 can be configured to be worn by the ambulatory patient 82 for at least several hours per day, and also during the night. That, for at least several days, even a few months. It will be understood that the support structure 170 is shown only generically in FIG. 1, and in fact partly conceptually. FIG. 1 is provided merely to illustrate concepts about the support structure 170, and is not to be construed as limiting how the support structure 170 is implemented, or how it is worn.

[0029] The support structure 170 can be implemented in many different ways. For example, it can be implemented in a single component or a combination of multiple components. In embodiments, the support structure 170 could include a vest, a half-vest, a garment, etc. In such embodiments such items can be worn similarly to analogous articles of clothing. In embodiments, the support structure 170 could include a harness, one or more belts or straps, etc. In such embodiments, such items can be worn by the patient around the torso, hips, over the shoulder, etc. In embodiments, the support structure 170 can include a container or housing, which can even be waterproof. In such embodiments, the support structure can be worn by being attached to the patient's body by adhesive material, for example as shown and described in U.S. Pat. No. 8,024,037. The support structure 170 can even be implemented as described for the support structure of US Pat. App. No. US2017/0056682, which is incorporated herein by reference. Of course, in such embodiments, the person skilled in the art will recognize that additional components of the WMS can be in the housing of a support structure instead of being attached externally to the support structure, for example as described in the US2017/0056682 document. There can be other examples.

[0030] FIG. 1 shows a sample external defibrillator and/or pacemaker 100. As described in more detail later in this document, some aspects of the external defibrillator and/or pacemaker 100 include a housing and an energy storage module within the housing. As such, in the context of a WMS, the defibrillator and/or pacemaker 100 is sometimes called a main electronics module. The energy storage module can be configured to store an electrical charge. Other components can cause at least some of the stored electrical charge to be discharged via electrodes through the patient, so as to deliver one or more electrical pulses or shocks through the patient. This action is also called shocking the patient.

[0031] FIG. 1 also shows sample electrotherapy electrodes 104, 108. The electrotherapy electrodes 104, 108 are also called therapy electrodes. When used for defibrillation, the electrotherapy electrodes 104, 108 are also called defibrillation electrodes. The therapy electrodes 104, 108 are

coupled to external defibrillator and/or pacer **100** via electrode leads **105**. The therapy electrodes **104**, **108** can be configured to be worn by the patient **82** in a number of ways. For instance, the defibrillator and/or pacer **100** and the therapy electrodes **104**, **108** can be coupled to the support structure **170**, directly or indirectly. In other words, the support structure **170** can be configured to be worn by the ambulatory patient **82** so as to maintain at least one of the electrodes **104**, **108** on the body of the ambulatory patient **82**, while the patient **82** is moving around, etc. The electrode can be thus maintained on the body by being attached to the skin of the patient **82**, simply pressed against the skin directly or through garments, etc. In some embodiments the electrode is not necessarily pressed against the skin, but becomes biased that way upon sensing a condition that could merit intervention by the WMS. In addition, many of the components of the defibrillator and/or pacer **100** can be considered coupled to the support structure **170** directly, or indirectly via at least one of the therapy electrodes **104**, **108**.

[0032] When the therapy electrodes **104**, **108** make good electrical contact with the body of the patient **82**, the defibrillator and/or pacer **100** can administer, via the therapy electrodes **104**, **108**, one or more brief electric pulses through the body of the patient **82**, such as defibrillation pulses and pacing pulses. These pulses are also known as electrotherapy and therapy, and have attributes suitable for their purpose.

[0033] A defibrillation pulse **111** is typically strong, having an energy of at least **100 Joule**. The defibrillation pulse **111** is also known as shock, defibrillation shock, therapy shock, etc. The defibrillation pulse **111** is intended to go through and restart the heart **85**, in an effort to save the life of the patient **82**.

[0034] A pacing pulse **121** is also shown, but it is not intended to be administered concurrently with the defibrillation pulse **111**. The pacing pulse **121** is intended to simply pace the heart **85** if needed, and typically a periodic sequence of pacing pulses is caused to be delivered by appropriately timed discharges. The pacing pulse **121** is shown as smaller than the defibrillation pulse **111** to reflect the fact that pacing pulses have less energy than defibrillation pulses, e.g. namely typically no more than **30 Joule**. In reality, the pacing pulse is a discharge from at least two therapy electrodes, which are either the same therapy electrodes **104**, **108** as for the defibrillation pulse **111**, or different therapy electrodes (not shown in FIG. 1).

[0035] A prior art defibrillator typically decides whether to defibrillate or not based on an ECG signal of the patient. However, the external defibrillator and/or pacer **100** may initiate defibrillation, or hold-off defibrillation, or may initiate pacing, based on a variety of inputs, with the ECG signal merely being one of these inputs.

[0036] A WMS according to embodiments can obtain data from the patient **82**. For collecting such data, the WMS may optionally include at least an outside monitoring device **180**. The device **180** is called an “outside” device because it could be provided as a standalone device, for example not within the housing of the defibrillator and/or pacer **100**. The device **180** can be configured to sense or monitor at least one local parameter. A local parameter can be a parameter of the patient **82**, or a parameter of the WMS, or a parameter of the environment, as will be described later in this document.

[0037] For some of these parameters, the device **180** may include one or more sensors or transducers. Each one of such

sensors can be configured to sense a parameter of the patient **82**, and to render an input responsive to the sensed parameter. In some embodiments the input is quantitative, such as values of a sensed parameter; in other embodiments the input is qualitative, such as informing whether or not a threshold is crossed, and so on. Sometimes these inputs about the patient **82** are also called physiological inputs and patient inputs. In embodiments, a sensor can be construed more broadly, as encompassing more than one individual sensors.

[0038] Optionally, the device **180** is physically coupled to the support structure **170**. In addition, the device **180** may be communicatively coupled with other components that are coupled to the support structure **170**. Such communication can be implemented by a communication module, as will be deemed applicable by a person skilled in the art in view of this description.

[0039] In embodiments, one or more of the components of the shown WMS may be customized for the patient **82**. This customization may include a number of aspects. For instance, the support structure **170** can be fitted to the body of the patient **82**. For another instance, baseline physiological parameters of the patient **82** can be measured for various scenarios, such as when the patient is lying down (various orientations), sitting, standing, walking, running, and so on. These baseline physiological parameters can be the heart rate of the patient **82**, motion detector outputs, one for each scenario, etc. The measured values of such baseline physiological parameters can be used to customize the WMS, in order to make its diagnoses more accurate, since patients' bodies differ from one another. Of course, such parameter values can be stored in a memory of the WMS, and so on. Moreover, a programming interface can be made according to embodiments, which receives such measured values of baseline physiological parameters. Such a programming interface may input automatically in the WMS these, along with other data.

[0040] FIG. 2 is a diagram showing components of an external defibrillator and/or pacer **200**, made according to embodiments. These components can be, for example, included in the external defibrillator and/or pacer **100** of FIG. 1. External defibrillator and/or pacer **200** is intended for a patient who would be carrying it on their body, such as ambulatory the patient **82** of FIG. 1. The components shown in FIG. 2 can be provided in a housing **201**, which may also be referred to as casing **201**.

[0041] The defibrillator and/or pacer **200** may include a user interface (UI) **280** for a user **282**. User **282** can be the patient **82**, also known as patient **282**, also known as the wearer **282**. Or, the user **282** can be a local rescuer at the scene, such as a bystander who might offer assistance, or a trained person. Or, the user **282** might be a remotely located trained caregiver in communication with the WMS.

[0042] The user interface **280** can be made in a number of ways. The user interface **280** may include output devices, which can be visual, audible or tactile, for communicating to a user by outputting images, sounds or vibrations. Images, sounds, vibrations, and anything that can be perceived by user **282** can also be called human-perceptible indications. As such, an output device according to embodiments can be configured to output a human-perceptible indication (HPI). Such HPIs can be used to alert the patient, sound alarms that may be intended also for bystanders, and so on. There are many examples of output devices. For example, an output

device can be a light that can be turned on and off, a screen to display what is sensed, detected and/or measured, and provide visual feedback to the local rescuer **282** for their resuscitation attempts, and so on. Another output device can be a speaker, which can be configured to issue voice prompts, alerts, beeps, loud alarm sounds and/or words, and so on. These can also be for bystanders, when defibrillating or just pacing, and so on.

[0043] The user interface **280** may further include input devices for receiving inputs from users. Such users can be the patient **82**, perhaps a local trained caregiver or a bystander, and so on. Such input devices may include various controls, such as pushbuttons, keyboards, touchscreens, one or more microphones, and so on. An input device can be a cancel switch, which is sometimes called an “I am alive” switch or “live man” switch. In some embodiments, actuating the cancel switch can prevent the impending delivery of a shock, or of pacing pulses. In particular, in some embodiments the speaker is configured to output a warning prompt prior to an impending or planned defibrillation shock or a pacing sequence of pacing pulses being caused to be delivered, and the cancel switch is configured to be actuated by the ambulatory patient **82** in response the warning prompt being output. In such embodiments, the impending or planned defibrillation shock or pacing sequence of the pacing pulses is not caused to be delivered responsive to the cancel switch having been actuated after the warning prompt has been output. Operations of the processor and methods may include causing the speaker to output a warning prompt prior to causing a (planned) pacing sequence of the pacing pulses to be delivered, and determining whether or not the cancel switch has been actuated after the warning prompt has been output.

[0044] The defibrillator and/or pacer **200** may include an internal monitoring device **281**. The device **281** is called an “internal” device because it is incorporated within the housing **201**. The monitoring device **281** can sense or monitor patient parameters such as patient physiological parameters, system parameters and/or environmental parameters, all of which can be called patient data. In other words, the internal monitoring device **281** can be complementary or an alternative to the outside monitoring device **180** of FIG. 1. Allocating which of the parameters are to be monitored by which of the monitoring devices **180**, **281** can be done according to design considerations. The device **281** may include one or more sensors, as also described elsewhere in this document.

[0045] Patient parameters may include patient physiological parameters. Patient physiological parameters may include, for example and without limitation, those physiological parameters that can be of any help in detecting by the WMS whether or not the patient is in need of a shock or other intervention or assistance. Patient physiological parameters may also optionally include the patient’s medical history, event history and so on. Examples of such parameters include the patient’s ECG, blood oxygen level, blood flow, blood pressure, blood perfusion, pulsatile change in light transmission or reflection properties of perfused tissue, heart sounds, heart wall motion, breathing sounds and pulse. Accordingly, the monitoring devices **180**, **281** may include one or more sensors configured to acquire patient physiological signals. Examples of such sensors or transducers include one or more electrodes to detect ECG data, a perfusion sensor, a pulse oximeter, a device for detecting

blood flow (e.g. a Doppler device), a sensor for detecting blood pressure (e.g. a cuff), an optical sensor, illumination detectors and sensors perhaps working together with light sources for detecting color change in tissue, a motion sensor, a device that can detect heart wall movement, a sound sensor, a device with a microphone, an SpO₂ sensor, and so on. In view of this disclosure, it will be appreciated that such sensors can help detect the patient’s pulse, and can therefore also be called pulse detection sensors, pulse sensors, and pulse rate sensors. In addition, a person skilled in the art may implement other ways of performing pulse detection.

[0046] In some embodiments, the local parameter is a trend that can be detected in a monitored physiological parameter of the patient **282**. A trend can be detected by comparing values of parameters at different times over short and long terms. Parameters whose detected trends can particularly help a cardiac rehabilitation program include: a) cardiac function (e.g. ejection fraction, stroke volume, cardiac output, etc.); b) heart rate variability at rest or during exercise; c) heart rate profile during exercise and measurement of activity vigor, such as from the profile of an accelerometer signal and informed from adaptive rate pacemaker technology; d) heart rate trending; e) perfusion, such as from SpO₂, CO₂, or other parameters such as those mentioned above, f) respiratory function, respiratory rate, etc.; g) motion, level of activity; and so on. Once a trend is detected, it can be stored and/or reported via a communication link, along perhaps with a warning if warranted. From the report, a physician monitoring the progress of the patient **282** will know about a condition that is either not improving or deteriorating.

[0047] Patient state parameters include recorded aspects of the patient **282**, such as motion, posture, whether they have spoken recently plus maybe also what they said, and so on, plus optionally the history of these parameters. Or, one of these monitoring devices could include a location sensor such as a Global Positioning System (GPS) location sensor. Such a sensor can detect the location, plus a speed can be detected as a rate of change of location over time. Many motion detectors output a motion signal that is indicative of the motion of the detector, and thus of the patient’s body. Patient state parameters can be very helpful in narrowing down the determination of whether SCA is indeed taking place.

[0048] A WMS made according to embodiments may thus include a motion detector. In embodiments, a motion detector can be implemented within the outside monitoring device **180** or within the internal monitoring device **281**. A motion detector of a WMS according to embodiments can be configured to detect a motion event. A motion event can be defined as is convenient, for example a change in motion from a baseline motion or rest, etc. In such cases, a sensed patient parameter is motion. Such a motion detector can be made in many ways as is known in the art, for example by using an accelerometer and so on. In this example, a motion detector **287** is implemented within the monitoring device **281**.

[0049] System parameters of a WMS can include system identification, battery status, system date and time, reports of self-testing, records of data entered, records of episodes and intervention, and so on. In response to the detected motion event, the motion detector may render or generate, from the detected motion event or motion, a motion detection input that can be received by a subsequent device or functionality.

[0050] Environmental parameters can include ambient temperature and pressure. Moreover, a humidity sensor may provide information as to whether or not it is likely raining. Presumed patient location could also be considered an environmental parameter. The patient location could be presumed, if the monitoring device 180 or 281 includes a GPS location sensor as per the above, and if it is presumed or sensed that the patient is wearing the WMS.

[0051] The defibrillator and/or pacer 200 typically includes a therapy delivery port 210. The therapy delivery port 210 can be a socket in the housing 201, or other equivalent structure. The therapy delivery port 210 includes electrical nodes 214, 218. Leads of the therapy electrodes 204, 208, such as the leads 105 of FIG. 1, can be plugged into the therapy delivery port 210, so as to make electrical contact with the nodes 214, 218, respectively. It is also possible that the therapy electrodes 204, 208 are connected continuously to the therapy delivery port 210, instead. Either way, the therapy delivery port 210 can be used for guiding, via electrodes, to the wearer at least some of the electrical charge that has been stored in an energy storage module 250 that is described more fully later in this document. The electric charge will be the shock for defibrillation, pacing, and so on.

[0052] The defibrillator and/or pacer 200 may optionally also have a sensor port 219 in the housing 201, which is also sometimes known as an ECG port. The sensor port 219 can be adapted for plugging in the sensing electrodes 209, which are also known as ECG electrodes and ECG leads. The ECG electrodes 209 in this example are distinct from the therapy electrodes 104, 108. It is also possible that the sensing electrodes 209 can be connected continuously to the sensor port 219, instead. The sensing electrodes 209 can be types of transducers that can help sense an ECG signal of the patient, e.g. a 12-lead signal, or a signal from a different number of leads, especially if they make good electrical contact with the body of the patient and in particular with the skin of the patient. As with the therapy electrodes 204, 208, the support structure can be configured to be worn by the patient 282 so as to maintain the sensing electrodes 209 on a body of the patient 282. For example, the sensing electrodes 209 can be attached to the inside of the support structure 170 for making good electrical contact with the patient, similarly with the therapy electrodes 204, 208.

[0053] Optionally a WMS according to embodiments also includes a fluid that it can deploy automatically between the electrodes and the patient's skin. The fluid can be conductive, such as by including an electrolyte, for establishing a better electrical contact between the electrodes and the skin. Electrically speaking, when the fluid is deployed, the electrical impedance between each electrode and the skin is reduced. Mechanically speaking, the fluid may be in the form of a low-viscosity gel. As such, it will not flow too far away from the location it is released, after being deployed. The fluid can be used for both the therapy electrodes 204, 208, and for the sensing electrodes 209.

[0054] The fluid may be initially stored in a fluid reservoir, not shown in FIG. 2. Such a fluid reservoir can be coupled to the support structure. In addition, a WMS according to embodiments further includes a fluid deploying mechanism 274. The fluid deploying mechanism 274 can be configured to cause at least some of the fluid to be released from the reservoir, and be deployed near one or both of the patient locations to which the therapy electrodes 204, 208 are

configured to be attached to the patient. In some embodiments, the fluid deploying mechanism 274 is activated prior to the electrical discharge responsive to receiving an activation signal AS from a processor 230, which is described more fully later in this document.

[0055] In some embodiments, defibrillator and/or pacer 200 also includes a measurement circuit 220, as one or more of its modules working together with its sensors or transducers. The measurement circuit 220 senses one or more electrical physiological signals of the patient from the sensor port 219, if provided. Even if the defibrillator and/or pacer 200 lacks a sensor port, the measurement circuit 220 may optionally obtain physiological signals through the nodes 214, 218 instead, when the therapy electrodes 204, 208 are attached to the patient. In these cases, the input reflects an ECG measurement. The patient parameter can be an ECG, which can be sensed as a voltage difference between electrodes 204, 208. In addition, the patient parameter can be an impedance (IMP. or Z), which can be sensed between the electrodes 204, 208 and/or between the connections of the sensor port 219 considered pairwise. Sensing the impedance can be useful for detecting, among other things, whether these electrodes 204, 208 and/or the sensing electrodes 209 are not making good electrical contact with the patient's body at the time. These patient physiological signals may be sensed when available. The measurement circuit 220 can then render or generate information about them as inputs, data, other signals, etc. As such, the measurement circuit 220 can be configured to render a patient input responsive to a patient parameter sensed by a sensor. In some embodiments, the measurement circuit 220 can be configured to render a patient input, such as values of an ECG signal, responsive to the ECG signal sensed by the sensing electrodes 209. More strictly speaking, the information rendered by the measurement circuit 220 is output from it, but this information can be called an input because it is received as an input by a subsequent stage, device or functionality.

[0056] The defibrillator and/or pacer 200 also includes a processor 230. The processor 230 may be implemented in a number of ways. Such ways include, by way of example and not of limitation, digital and/or analog processors such as microprocessors and Digital Signal Processors (DSPs), controllers such as microcontrollers, software running in a machine, programmable circuits such as Field Programmable Gate Arrays (FPGAs), Field-Programmable Analog Arrays (FPAAs), Programmable Logic Devices (PLDs), Application Specific Integrated Circuits (ASICs). In various embodiments, the processor may be implemented using any combination of one or more of the above types of processors, and so on. In some embodiments, the processor may be implemented using multiple electronic devices distributed in various parts of external defibrillator and/or pacer 200.

[0057] The processor 230 may include, or have access to, a non-transitory storage medium, such as a memory 238 that is described more fully later in this document. Such a memory can have a non-volatile component for storage of machine-readable and machine-executable instructions. A set of such instructions can also be called a program. The instructions, which may also be referred to as "software," generally provide functionality by performing acts, operations and/or methods as may be disclosed herein or understood by one skilled in the art in view of the disclosed embodiments. In some embodiments, and as a matter of convention used herein, instances of the software may be

referred to as a “module” and by other similar terms. Generally, a module includes a set of the instructions so as to offer or fulfill a particular functionality. Embodiments of modules and the functionality delivered are not limited by the embodiments described in this document.

[0058] The processor **230** can be considered to have a number of modules. One such module can be a detection module **232**. The detection module **232** can include a Ventricular Fibrillation (VF) detector. The patient’s sensed ECG from measurement circuit **220**, which can be available as inputs, data that reflect values, or values of other signals, may be used by the VF detector to determine whether the patient is experiencing VF. Detecting VF is useful, because VF typically results in SCA. The detection module **232** can also include a Ventricular Tachycardia (VT) detector for detecting VT, and so on.

[0059] Another such module in processor **230** can be an advice module **234**, which generates advice for what to do. The advice can be based on outputs of the detection module **232**. There can be many types of advice according to embodiments. In some embodiments, the advice is a shock/no shock determination that processor **230** can make, for example via advice module **234**. The shock/no shock determination can be made by executing a stored Shock Advisory Algorithm. A Shock Advisory Algorithm can make a shock/no shock determination from one or more ECG signals that are captured according to embodiments, and determine whether or not a shock criterion is met. The determination can be made from a rhythm analysis of the captured ECG signal or otherwise. For example, there can be shock decisions for VF, VT, etc.

[0060] In some embodiments, when the determination is to shock, an electrical charge is delivered to the patient. Delivering the electrical charge is also known as discharging and shocking the patient. As mentioned above, such can be for defibrillation, pacing, and so on.

[0061] In perfect conditions, a very reliable shock/no shock determination can be made from a segment of the sensed ECG signal of the patient. In practice, however, the ECG signal is often corrupted by electrical noise, which makes it difficult to analyze. Too much noise sometimes causes an incorrect detection of a heart arrhythmia, resulting in a false alarm to the patient. Noisy ECG signals may be handled as described in published US patent application No. US 2019/0030351 A1, and No. US 2019/0030352 A1, and which are incorporated herein by reference.

[0062] The processor **230** can include additional modules, such as other module **236**, for other functions. In addition, if the internal monitoring device **281** is indeed provided, the processor **230** may receive its inputs, etc.

[0063] The defibrillator and/or pacer **200** optionally further includes a memory **238**, which can work together with the processor **230**. The memory **238** may be implemented in a number of ways. Such ways include, by way of example and not of limitation, volatile memories, Non-volatile Memories (NVM), Read-Only Memories (ROM), Random Access Memories (RAM), magnetic disk storage media, optical storage media, smart cards, flash memory devices, any combination of these, and so on. The memory **238** is thus a non-transitory storage medium. The memory **238**, if provided, can include programs for the processor **230**, which the processor **230** may be able to read and execute. More particularly, the programs can include sets of instructions in the form of code, which the processor **230** may be able to

execute upon reading. Executing is performed by physical manipulations of physical quantities, and may result in functions, operations, processes, acts, actions and/or methods to be performed, and/or the processor to cause other devices or components or blocks to perform such functions, operations, processes, acts, actions and/or methods. The programs can be operational for the inherent needs of the processor **230**, and can also include protocols and ways that decisions can be made by the advice module **234**. In addition, the memory **238** can store prompts for the user **282**, if this user is a local rescuer. Moreover, the memory **238** can store data. This data can include patient data, system data and environmental data, for example as learned by the internal monitoring device **281** and the outside monitoring device **180**. The data can be stored in the memory **238** before it is transmitted out of defibrillator and/or pacer **200**, or be stored there after it is received by the defibrillator and/or pacer **200**.

[0064] The defibrillator and/or pacer **200** can optionally include a communication module **290**, for establishing one or more wired or wireless communication links with other devices of other entities, such as a remote assistance center, Emergency Medical Services (EMS), and so on. The communication links can be used to transfer data and commands. The data may be patient data, event information, therapy attempted, CPR performance, system data, environmental data, and so on. For example, the communication module **290** may transmit wirelessly, e.g. on a daily basis, heart rate, respiratory rate, and other vital signs data to a server accessible over the internet, for instance as described in US 20140043149. This data can be analyzed directly by the patient’s physician and can also be analyzed automatically by algorithms designed to detect a developing illness and then notify medical personnel via text, email, phone, etc. The module **290** may also include such interconnected sub-components as may be deemed necessary by a person skilled in the art, for example an antenna, portions of a processor, supporting electronics, outlet for a telephone or a network cable, etc.

[0065] The defibrillator and/or pacer **200** may also include a power source **240**, which is configured to provide electrical charge in the form of a current. To enable portability of the defibrillator and/or pacer **200**, the power source **240** typically includes a battery. Such a battery is typically implemented as a battery pack, which can be rechargeable or not. Sometimes a combination is used of rechargeable and non-rechargeable battery packs. Other embodiments of the power source **240** can include an AC power override, for where AC power will be available, an energy-storing capacitor, and so on. Appropriate components may be included to provide for charging or replacing the power source **240**. In some embodiments, the power source **240** is controlled and/or monitored by the processor **230**.

[0066] The defibrillator and/or pacer **200** may additionally include an energy storage module **250**. The energy storage module **250** can be coupled to the support structure of the WMS, for example either directly or via the electrodes and their leads. The energy storage module **250** can be coupled to receive the electrical charge provided by the power source **240**. The energy storage module **250** can be configured to store the electrical charge received by the power source. As such, the energy storage module **250** is where some electrical energy can be stored temporarily in the form of an electrical charge, when preparing it for discharge to admin-

ister a shock. In embodiments, the module **250** can be charged from the power source **240** to the desired amount of energy, for instance as controlled by the processor **230**. In typical implementations, the module **250** includes a capacitor **252**, which can be a single capacitor or a system of capacitors, and so on. In some embodiments, the energy storage module **250** includes a device that exhibits high power density, such as an ultracapacitor. As described above, the capacitor **252** can store the energy in the form of an electrical charge, for delivering to the patient.

[0067] A decision to shock can be made responsive to the shock criterion being met, as per the above-mentioned determination. When the decision is to shock, the processor **230** can be configured to cause at least some or all of the electrical charge stored in the module **250** to be discharged through the patient **82** while the support structure is worn by the patient **82**, so as to deliver the shock **111** to the patient **82**.

[0068] For causing the discharge, the defibrillator and/or pacer **200** moreover includes a discharge circuit **255**. When the decision is to shock, the processor **230** can be configured to control the discharge circuit **255** to discharge through the patient at least some of all of the electrical charge stored in the energy storage module **250**, especially in a desired waveform. When the decision is to merely pace, i.e., to deliver pacing pulses, the processor **230** can be configured to cause control the discharge circuit **255** to discharge through the patient at least some of the electrical charge provided by the power source **240**. Since pacing requires lesser charge and/or energy than a defibrillation shock, in some embodiments pacing wiring **241** is provided from the power source **240** to the discharge circuit **255**. The pacing wiring **241** is shown as two wires that bypass the energy storage module **250**. As such, the energy for the pacing is provided by the power source **240** either directly via the pacing wiring **241**, or through the discharge circuit **250**. And, in some embodiments where only a pacer is provided, the energy storage module **250** is not needed, and enough pacing current may be provided from the power source **240**. Either way, discharging can be to the nodes **214**, **218**, and from there to the therapy electrodes **204**, **208**, so as to cause a shock to be delivered to the patient. The circuit **255** can include one or more switches **257**. The switches **257** can be made in a number of ways, such as by an H-bridge, and so on. In some embodiments, different ones of the switches **257** may be used for a discharge where a defibrillation shock is caused to be delivered, than for a discharge where the much weaker pacing pulses are caused to be delivered. The circuit **255** could also be thus controlled via the processor **230**, and/or the user interface **280**.

[0069] The pacing capability can be implemented in a number of ways. ECG sensing may be done in the processor, as mentioned elsewhere in this document, or separately, for demand or synchronous pacing. In some embodiments, however, pacing can be asynchronous. Pacing can be software controlled, e.g., by managing the defibrillation path, or a separate pacing therapy circuit (not shown) could be included, which can receive the ECG sensing.

[0070] A time waveform of the discharge may be controlled by thus controlling discharge circuit **255**. The amount of energy of the discharge can be controlled by how much energy storage module has been charged, and also by how long the discharge circuit **255** is controlled to remain open.

[0071] The defibrillator and/or pacer **200** can optionally include other components.

[0072] FIG. **3** is a diagram of sample embodiments of components of an WMS. A support structure **370** includes a vest-like wearable garment. The support structure **370** has a back side **371**, and a front side **372** that closes in front of the chest of the patient.

[0073] The WMS of FIG. **3** also includes an external defibrillator and/or pacer **300**. FIG. **3** does not show any support for external defibrillator and/or pacer **300**, which may be carried in a purse, on a belt, by a strap over the shoulder, on the support structure **370**, and so on. Wires **305** connect the external defibrillator and/or pacer **300** to electrodes **304**, **308**, **309**. Of those, the electrodes **304**, **308** are therapy electrodes, and the electrodes **309** are ECG sensing electrodes. The therapy electrodes **104**, **108**, **204**, **208**, **304**, **308** can also be called discharge electrodes.

[0074] The support structure **370** is configured to be worn by the ambulatory patient so as to maintain the electrodes **304**, **308**, **309** on a body of the patient. Indeed, the back therapy electrodes **308** are maintained in pockets **378**. Of course, the inside of the pockets **378** can be made with loose netting, so that the electrodes **308** can contact the back of the patient, especially with the help of the conductive fluid that has been deployed. In addition, the sensing electrodes **309** are maintained in positions that surround the patient's torso, for sensing ECG signals and/or the impedance of the patient.

[0075] ECG signals in a WMS may include too much electrical noise to be useful. To ameliorate the problem, multiple ECG sensing electrodes **309** are provided, for presenting many options to the processor **230**. These options are different vectors for sensing the ECG signal, as described now in more detail.

[0076] WMSs according to embodiments advantageously also include one or more pacing capabilities. For pacing, instead of defibrillating the patient with a strong defibrillation pulse, at least some of the stored electrical charge is caused to be discharged via at least two of the therapy electrodes **104**, **108**, **204**, **208**, **304**, **308** through the ambulatory patient **82**, so as to deliver to the ambulatory patient **82** a pacing sequence of pacing pulses. In other words, the pacing pulses may be delivered via the therapy electrodes. The pacing pulses may have certain energies, respectively. The energies may be less than 30 Joule, and usually about 10 J. The pacing pulses may have certain spacings between successive ones of them. These spacings define the pacing period, and thus the pacing rate of the pacing sequence. The pacing pulses may be periodic, but there is no requirement for them to be so.

[0077] In embodiments, the WMS may detect when the patient's heart rhythm starts to deteriorate, but before the rhythm reaches the state where the patient needs to be defibrillated. In such embodiments, the WMS may pace the patient first, and hopefully not have to resort to the full intervention of defibrillation. Of course, if the patient does not respond to the pacing, the WMS may then later cause a defibrillation shock to be delivered.

[0078] In some embodiments, the pacing is controlled by the processor **230**. It is the processor **230** that can analyze the ECG signal, or portions of it, that are sensed by the ECG electrodes. The analysis can be to determine whether or not the ECG signal meets one or more criteria, which is also known as analyzing the ECG signal against these criteria. The analysis by the processor **230** can also be for operations

shown in the flowcharts of this document, including compute a heart rate, turn on and off QRS detection, and so on. Responsive to the analysis, the processor 230 can be configured to cause pacing pulses to be delivered, with the desired energy, spacings between successive ones of and waveform. The processor 230 can be further configured to change the pacing pulses, stop them, defibrillate, and so on, in addition to all other functions such as cause communication and so on.

[0079] In some embodiments, a WMS detects a bradycardia of one or more types, and then confirms the detection, before pacing to treat the detected bradycardia. The detected bradycardia here can be heart arrhythmias where the heart is beating spontaneously more slowly than normal, or not at all, and include at least bradycardia, extreme bradycardia and asystole. Examples are now described.

[0080] FIG. 4 shows a flowchart 400 for describing methods according to embodiments. In addition, FIG. 4 shows a time diagram of a sample ECG signal 404 along a time axis 409. Also, FIG. 4 shows a small fraction of a sample pacing sequence 491, in the form of a timing diagram. Moreover, FIG. 4 shows an icon of a defibrillation shock 411, similar to the defibrillation shock 111 of FIG. 1.

[0081] The method of flowchart 400 can start at an operation 410, where an ECG signal 404 of an ambulatory patient can be sensed.

[0082] According to another operation 433, a first portion 401 of the sensed ECG signal 404 can be analyzed against a first bradycardia criterion. The first bradycardia criterion may be implemented in a number of ways. In some embodiments, the first bradycardia criterion includes having detected, in the first portion 401 of the ECG signal 404, heartbeats that have a heart rate of less than 40 beats per minute (bpm). Or, less than 30 bpm. Or, less than 20 bpm for extreme bradycardia, and so on. Notably, when the bradycardia criterion includes having detected a heart rate of less than 20 bpm, that could independently be also a criterion for asystole. In embodiments, therefore, asystole may be detected as well.

[0083] For better explanation of bradycardia criteria, the waveform 403 is an idealized version of what ECG signal 404 might be, if the patient were perfectly healthy, at a normal time, and with no noise present. The analysis attempts to detect, and here would find, three R peaks R1, R2, R3. Between two successive ones of them, these three R peaks R1, R2, R3 have spacings PR. These spacings need not be identical, of course, but they usually are substantially equal. The average of these spacings would correspond to the heart rate, which then would be applied to the criterion.

[0084] According to another operation 434, which is shown as a decision diamond 434, it can be determined whether or not the first bradycardia criterion is met at the operation 433. If not then, at another operation 480, a defibrillation shock 411 may be later delivered, in embodiments where a defibrillator is also provided. It can be delivered later, because some time may pass with other events, and optional other operations 470 may be performed in the interim, such as other diagnoses, prompts, and treatments. For instance, at the operation 470, a defibrillation criterion may be first determined to be met. The defibrillation criterion may be met later, and not in connection to any bradycardia. For performing the operation 480, the processor can be configured to cause at least some of the stored electrical charge to be discharged via at least two of the

therapy electrodes through the ambulatory patient, so as to deliver to the ambulatory patient a defibrillation shock 411 having an energy of at least 100 Joule.

[0085] If, at the operation 434, it is determined that the first bradycardia criterion of the operation 433 is met then, according to another operation 435, there can be waiting, for a confirmation period 408 of at least 10 sec. The confirmation period 408 can last longer, for example at least 20 sec, at least 50 sec such as 60 sec, and so on. Again, the waiting for the confirmation period can be performed responsive to the first portion 401 of the analyzed ECG signal 404 meeting the first bradycardia criterion of the operation 433.

[0086] According to another operation 436, a second portion 402 of the sensed ECG signal 404 may be analyzed. At least a fraction of the second portion 402 of the sensed ECG signal 404 may have been sensed during the confirmation period 408. Or, even the entire second portion 402 of the sensed ECG signal 404 may have been sensed during the confirmation period 408.

[0087] At the operation 436 the second portion 402 may be analyzed against a second bradycardia criterion. The second bradycardia criterion can be the same, or different, as the first bradycardia criterion, of course as applied to the second portion 402 of the ECG signal instead of to the first portion 401.

[0088] According to another operation 437, it can be determined whether or not the second bradycardia criterion is met at the operation 436. If not then, then execution may proceed to the operation 480, similarly with a no answer from the operation 434.

[0089] If, at the operation 437, it is determined that the second bradycardia criterion of the operation 436 is met, then another operation 490 may be performed. Prior to the operation 490, according to an optional operation 488, the ambulatory patient 82 may be warned of the impending pacing. In particular, a speaker of the UI 280 may output a warning prompt prior to the pacing sequence of the pacing pulses of the operation 490 being caused to be delivered. According to an optional operation 489, the processor can be further configured to determine whether or not the cancel switch has been actuated after the warning prompt has been output. In such embodiments, the pacing sequence of the pacing pulses is not caused to be delivered responsive to determining that the cancel switch has been actuated after the warning prompt has been output. The operation 488 may be performed at different times prior to the operation 488; in particular, the warning prompt can be output prior to, concurrently with, or after analyzing the second portion of the sensed ECG signal against the second bradycardia criterion, and so on. In this diagram, the operation 488 is shown as being performed after the operation 437, but that is only a non-limiting example.

[0090] Then, according to another operation 490, at least some of the electrical charge provided by the source can be caused to be discharged via at least two of the therapy electrodes through the ambulatory patient so as to deliver to the ambulatory patient a pacing sequence 491 of pacing pulses 422, 423, 424. The pacing pulses 422, 423, 424 may have certain energies, at least some of the certain energies being at most 30 Joule, as per the above for pacing pulses. The energy storage module 252 may be recharged after one or more pacing pulses. Again, the pacing sequence 491 can be caused to be delivered responsive to the second portion 402 of the analyzed ECG signal 404 meeting the second

bradycardia criterion of the operation 437. In embodiments, low-rate (e.g., 40 bpm), life-sustaining, pacing therapy can be thus provided in response to sustained, extremely low intrinsic heart rates (e.g., <30 bpm) or extended periods of asystole.

[0091] The pacing pulses 422, 423, 424 may have certain spacings PACP between them, which are also known as pacing periods. These may be equal to each other, or not. In some embodiments, the certain spacings PACP between the pacing pulses 422, 423, 424 of the pacing sequence 491 amount to a pacing rate having a value between 30 beats per minute (bpm) and 90 bpm. Similarly a group of, say, 10 such pacing pulses taken together may have 9 spacings among them, the 9 spacings having an average spacing that amounts to such a pacing rate.

[0092] According to another, optional operation 493, the pacing sequence 491 has a certain duration, which means it ends at some point. In such embodiments, the processor can be further configured to analyze, after the certain duration, a third portion of the sensed ECG signal, which is sensed after the second portion 402 of the sensed ECG signal 404.

[0093] Responsive to a result of the analysis of the operation 493, a number of operations may be performed. In some instances, the defibrillation shock of operation 480 is caused to be delivered, for instance to terminate any starting ventricular tachycardia (VT) or ventricular fibrillation (VF). In other instances, pacing stops, and other examples will be described below with reference to FIG. 8. In other instances, such as if bradycardia is detected, pacing resumes, to deliver to the patient another pacing sequence of pacing pulses of the same or different parameters. Plus, this sequence of “pace, stop, sense ECG, analyze ECG, then pace again if needed” may also be performed without the waiting for the confirmation period.

[0094] In other embodiments, at the operation 493, the third portion is analyzed against an escape pacing criterion. The escape pacing criterion may include having detected, in the third portion of the ECG signal, heartbeats that have a heart rate of at least 30 beats per minute (bpm). Or, at least 40 bpm. In such embodiments, responsive to the third portion of the analyzed ECG signal meeting the escape pacing criterion, the processor might not cause any of the electrical charge provided by the source to be discharged through the ambulatory patient for at least 60 sec, or even longer.

[0095] It will be appreciated that, when the sample ECG signal 404 is sensed in some of the embodiments of FIG. 4, no pacing is taking place. As such, the sample ECG signal 404 does not include any artifacts, such as little spikes that have been added from the pacing.

[0096] This is not always the case with embodiments. In some embodiments, the ECG signal is sensed while pacing is applied. Sample ECG signals are now shown and described, where pacing markers have been added to shown when pacing took place.

[0097] FIG. 5 is a diagram of a waveform of a sample actual ECG signal 504, which is sensed while the patient is being paced according to embodiments. In this example there is no appreciable noise.

[0098] In FIG. 5, pacing markers 522, 523, 524, 525, 526 can be discerned easily thanks to the lack of noise. It will be appreciated that, in this example, the pacing has likely captured the heart, which means that the pacing markers 522, 523, 524, 525, 526 actually drive corresponding heart-

beats, whose T waves are denoted as 572, 573, 574, 575, 576. The R waves that take place before these T waves are obscured by the pacer refractory period, and are thus not visible in FIG. 5. Such capture does not happen always.

[0099] One more observation about the pacing rate in FIG. 5 is that the pacing is relatively fast. Each time, the next pacing pulse is delivered very soon after the previous driven heartbeat. This does not allow any opportunity for intrinsic heartbeats, for reasons that are explained later in this document.

[0100] FIG. 6 is a diagram of a waveform of a sample actual ECG signal 604, which is sensed while the patient is being paced according to embodiments. In this example there is appreciable noise, as would be the case with the ambulatory patient being conscious and moving.

[0101] In FIG. 6, pacing markers 622, 623, 624, 625 are pointed out. There are large noise artifacts 699. Here two intrinsic heartbeats 673, 674 are detected, and this is an example of pacing with no capture.

[0102] From FIG. 6, the deleterious effects of electrical noise can be appreciated. Indeed, electrical noise in the ECG signal may cause detection to not detect heartbeats, or detect heartbeats where there are none, and accordingly misdiagnose the condition of the patient, with a potentially deadly outcome.

[0103] In some embodiments, a WMS applies the same pacing sequence to treat bradycardia and asystole. Optionally, it also applies this pacing sequence after a defibrillation shock. Examples are now described.

[0104] FIG. 7 shows a flowchart 700 for describing methods according to embodiments. In addition, FIG. 7 shows a sample ECG signal 704, and a small idealized ECG waveform 702 for explaining a bradycardia criterion. Also, FIG. 7 shows a small fraction of a sample pacing sequence 791 in the form of a timing diagram. Moreover, FIG. 7 shows an icon of a defibrillation shock 711, similar to those of the defibrillation shocks 111, 411.

[0105] The method of flowchart 700 can start at an operation 710, where an ECG signal 704 of an ambulatory patient can be sensed. The ECG signal 704 can be as described earlier, for instance for ECG signal 404.

[0106] According to another operation 730, the sensed ECG signal 704 can be analyzed against a bradycardia criterion, to detect whether the patient is suffering from bradycardia. The bradycardia criterion may be implemented in a number of ways. For instance, the bradycardia criterion may be that a detected heart rate is less than 50 bpm, but also larger than 15 bpm. Less than 15 bpm might be bradycardia or asystole. The heartbeats may be detected from their most prominent feature, namely the R peaks of QRS complexes. Again, the waveform 702 is an idealized version of what ECG signal 704 might be; in this example, the waveform 702 is also identical to the waveform 403, and so on.

[0107] According to another operation 731, it can be determined whether or not the bradycardia criterion is met at the operation 730. If, at the operation 731, it is determined that the bradycardia criterion of the operation 730 is met, then another operation 790 may be performed. Prior to the operation 790, according to an optional operation 788, the ambulatory patient 82 may be warned of the impending pacing. In particular, a speaker of the UI 280 may output a warning prompt prior to the planned pacing sequence of the pacing pulses of the operation 790 being caused to be delivered. According to an optional operation 789, the

processor can be further configured to determine whether or not the cancel switch has been actuated after the warning prompt has been output. In such embodiments, the planned pacing sequence of the pacing pulses is not caused to be delivered responsive to determining that the cancel switch has been actuated after the warning prompt has been output.

[0108] Then, according to another operation 790, a pacing sequence may be delivered to the patient. An example is shown in the sample short pacing sequence 791, with pacing pulses 722, 723, 724, and which can be similar to the short pacing sequence 491 of FIG. 4. The pacing sequence 791 may be periodic or not, demand pacing or not. In particular, for the operation 790, the processor can be configured to cause at least some of the electrical charge provided by the source to be discharged via at least two of the therapy electrodes through the ambulatory patient so as to deliver to the ambulatory patient a planned pacing sequence of pacing pulses. The pacing pulses may have certain energies and successive ones of certain spacings between them. In this example, the certain spacings are indicated as PACP. At least some of the certain energies are at most 30 Joule. The pacing sequence is planned, in that it has certain energies, and timings, which define the spacings. The pacing sequence can further be planned by designating how many pacing pulses will be delivered, before stopping and looking.

[0109] The pacing pulses of the planned pacing sequence have waveforms. These waveforms can be monophasic, biphasic, and so on. In some embodiments, the waveforms are the same regardless of whether they are due to the bradycardia criterion being met, or the asystole criterion being met. For instance, at least one of the pacing pulses of the planned pacing sequence has a certain waveform when delivered responsive to the sensed ECG signal meeting the bradycardia criterion, and at least one of the pacing pulses of the planned pacing sequence can have the certain waveform when delivered responsive to the sensed ECG signal meeting the asystole criterion. Similarly, at least one of the pacing pulses of the planned pacing sequence can have the certain waveform when delivered after the defibrillation shock has been caused to be delivered per the operation 780.

[0110] If, at the operation 731, it is determined that the bradycardia criterion of the operation 730 is not met then, according to another operation 750, the sensed ECG signal 704 can be analyzed against an asystole criterion, to detect whether the patient is suffering from asystole. Asystole is commonly known as flatlining, and corresponds to no electrical activity by the heart. The asystole criterion may be implemented in a number of ways, and can be distinct from the bradycardia criterion. In some embodiments, the asystole criterion includes detecting, in the ECG signal, no electrical activity that is larger than 200 micro Volt peak-to-peak. 100 micro Volt may be a good value. This detection can be for a suitable duration, such as at least 1.5 sec, at least 3 sec, at least 6 sec, perhaps 10 sec, and so on. For such detection noise may be removed, for example by cross-referencing to other channels, and so on.

[0111] According to another operation 751, it can be determined whether or not the asystole criterion is met at the operation 750. If, at the operation 751, it is determined that the asystole criterion of the operation 750 is met, then execution may proceed again to the operation 790, according to which the pacing sequence is delivered to the patient.

[0112] If, at the operation 751, it is determined that the asystole criterion of the operation 750 is not met then,

according to another operation 780, a defibrillation shock 711 may be later delivered, in embodiments where a defibrillator is also provided, similarly with what was described for operation 480. It can be later, because some time may pass with other events, and optional other operations 770 may be performed in the interim, such as other diagnoses, prompts, and treatments.

[0113] Optionally, after operation 780, the operation 790 may be performed.

[0114] In some embodiments, a WMS paces the patient while concurrently monitoring for arrhythmias, and it may continue, adjust, or end the pacing in response to the monitoring. These arrhythmias may include different types of bradyarrhythmia and of tachyarrhythmia. An advantage is that, while external pacing can sometimes be pro-arrhythmic (in fact, if an intrinsic beat is not detected and a pacing pulse is applied on the T wave, then VF may be triggered), with the monitoring of embodiments, it can be detected if the patient has flipped into VF which, if left untreated, is lethal. Pacing is not effective during VF. A much larger defibrillation shock is necessary to terminate VF. This method is not guaranteed to sense every intrinsic beat that a patient may generate, but if there is a sustained rate recovery it will be detected. It is certainly better than open-loop pacing, which would provide no sensing capability at all. Examples are now described.

[0115] FIG. 8 shows a number of elements. A sample planned sequence for pacing 820-A is shown as a time diagram. Sample pacing pulses . . . , 822, 823, 824, 825, which have started before pacing pulse 822, are shown against a time axis 829-A. In this example, the four sample pacing pulses 822, 823, 824, 825 are shown as occurring periodically, i.e., at regular intervals, with a spacing PACP1. And, since the spacing PACP1 is along a time axis, the spacing PACP1 is a time duration.

[0116] In FIG. 8 the four sample pacing pulses 822, 823, 824, 825 are shown as occurring periodically, but that is only for convenience of explanation and not necessary. In embodiments, a planned sequence 820-A need not be periodic, which means that the spacings between any two successive ones of these pacing pulses need not be identical. In some embodiments, at least some of the pacing pulses of the planned sequence can be delivered substantially periodically. For instance, in some embodiments, the planned sequence includes that the pacing pulses are delivered in successive groups that are separated by pauses. An example is now described.

[0117] FIG. 9 shows a sample planned sequence for pacing 920. Sample groups . . . , 971, 972, 973, 974, . . . of pacing pulses are shown as a time diagram, plotted against a time axis 929. Within each of the groups 971, 972, 973, 974, the pacing pulses are actually periodic with a period PG, although that is not necessary either. Since they are periodic, these groups 971, 972, 973, 974 can also be called periodic groups.

[0118] Between these groups 971, 972, 973, 974 there are pauses 984, with period PG. For instance, the left-most shown of these pauses 984 is between a last one 924 of the pacing pulses in group 971, and a first one 925 of the pacing pulses in the next group 972. As will be seen later in this document, these pauses 984 are longer, meaning PS is larger than PG, to present a better opportunity to detect intrinsic heartbeats.

[0119] It will be further appreciated that the planned sequence 920 is itself periodic, if one considers that the repeating pattern includes one of the groups and its following pause.

[0120] In embodiments, pulses are delivered that do not have many interruptions. Or, the interruptions are minimized. For instance, a planned sequence may have at least 5 successive pacing pulses with four spacings between successive ones of them. At the same time, no two of the 5 successive pacing pulses have a spacing between them that is longer than 4 times an average of the four spacings. Or 3 times. Or 2 times. Or 1.7 times. Or 1.4 times. The smaller this number, the better the heart is covered with pacing. A brief periodic pause in pacing should not significantly impact pacing efficacy. One 5 second pause every 30 seconds still means that pacing pulses are delivered 83% of the time. One 5 second pause every 60 seconds would result in pacing pulses 92% of the time. Either one of these options is much better than having no pacing pulses at all. Having a higher pacing duty cycle would be preferable, but that needs to be weighed against the tradeoff of being able to see the patient's intrinsic rhythm less often.

[0121] For instance, in the example of FIG. 9, the following 5 successive pacing pulses may be considered: pulse 924 plus all of the pulses of group 972. Their four spacings are PS, PG, PG, PG. The average duration AVE_PER of these four spacings would be given by Equation 1.

$$\text{AVE_PER} = [PS + PG + PG + PG] / 4 \quad \text{Equation (1)}$$

[0122] And none of the spacings, for example the largest spacing PS would be longer than 4 times AVE_PER, or 3 times, or 2 times, or 1.7 times, or 1.4 times per the above. This Equation (1), and what it stands for, is shown with reference to the example of FIG. 9, but that is not necessary, and it can apply to other planned sequences 820-A. If $PS = 2 \times PG$, then the pattern is to simply skip a pacing pulse.

[0123] Returning to FIG. 8, a sample ECG signal 804 is shown, along the time axis 829-A of the planned sequence 820-A. Specifically, a particular portion 812 of the ECG signal 804 has been sensed after the delivery per the planned sequence starts, and before it stops. As applied to this example, this means that the particular portion 812 has been sensed after the pacing pulse 822 has been delivered, and before the pacing pulse 825 has been delivered.

[0124] FIG. 8 thus shows how particular portion 812 of the ECG signal 804 is detected concurrently with the pacing of the planned sequence 820-A. The particular portion 812 is then analyzed and, according to a decision diamond 833, a continuance determination is made of whether or not to continue pacing according to the planned sequence 820-A.

[0125] Regarding the analyzing of the ECG signal, in some embodiments it is attempted to detect all heartbeats, in order to determine how successful the pacing is in: a) capturing the heart in the first place and, hopefully in b) eliciting intrinsic heartbeats. In other embodiments, or at other times, only intrinsic heartbeats are detected by a special detection technique that is now described.

[0126] FIG. 10 shows a planned sequence 1020-A against a time axis 1029-A. The planned sequence 1020-A includes pacing pulses . . . , 1024, 1025, The pacing pulse 1024 occurs at a time TM1 and the pacing pulse 1025 occurs at a

time TM3. The pacing pulses 1024, 1025 could be any two successive pulses of the previously described sequences 820-A or 920. In fact, they could be the pulses 924, 925, in which case the time distance, or spacing, between TM1 and TM3 would be PS.

[0127] A time TM2 divides the spacing between the two pacing pulses 1024, 1025 into an Interval A' 1081 and Interval B' 1082. An idealized ECG signal 1004 shows two different types of detected heartbeats.

[0128] In this example, during Interval A' 1081, a heartbeat 1072 occurs, and is considered to be a driven heartbeat. It is driven, in the sense that it was caused to happen by the pacing pulse 1024. This could be part of demand pacing. A heartbeat might not always be caused to happen by a pacing pulse. Even if the heartbeat 1072 occurs, it may not be detected. And, even if the heartbeat 1072 could be detected, it is not always detected. In some embodiments, detection of the driven heartbeat is not performed; rather, the delivery of the pacing pulse 1024 is assumed to capture the heart and detection during the interval 1081 is not required. The pacing pulse 1024 defines the beginning of the interval 1082 at TM2, at which point detection starts.

[0129] In this example, during Interval B' 1082, a heartbeat 1091 is detected, and is considered to be an intrinsic heartbeat. It is intrinsic, in the sense that it was not directly caused by the pacing pulse 1024, as was the driven heartbeat 1072. Rather, the heart by itself generated this intrinsic heartbeat 1091, perhaps due to its conditioning from previous pacing. An intrinsic heartbeat might not always be detected. Detecting an intrinsic heart beat may inhibit pace pulse delivery, as known for demand pacing operations.

[0130] In some embodiments, detection techniques involve turning off and on the QRS detection. This may involve turning ON and OFF a QRS detector, or alternate between considering the output of a QRS detector and disregarding or ignoring that output. Such alternating QRS detection is shown in a time diagram 1008. A vertical axis 1007 has ON and OFF values, and is perpendicular to a time axis 1009-A. The time axis 1009-A runs concurrently with the time axis 1029-A, as shown by the long perpendicular lines 1001.

[0131] For purposes of running QRS detection, intervals are now defined. The Interval B' 1082 can be thought of as an active period that starts after one of the pacing pulses 1024 is delivered, meaning after time TM1, and ends in conjunction with the next one of the pacing pulses 1025 being delivered, meaning in conjunction with time TM3. Here, "in conjunction" means that the end of the Interval B' 1082 is planned that way, and could be somewhat before the time TM3, or right on the time TM3, and so on. The QRS detector can be ON for the active period P_ACTIVE.

[0132] The Interval A' 1081 can be thought of as a pacer refractory period that starts in conjunction with the one of the pacing pulses 1024 being delivered, meaning in conjunction with time TM1, and ends when the above-described active period starts, meaning at time TM2. This pacer refractory period can be complementary to the active period. Here, "in conjunction" means that the start of the Interval A' 1081 is planned that way, and could be somewhat after the time TM1, or right on the time TM1, and so on. The QRS detection can be OFF for the pacer refractory period P_PACER REFRACTORY. The pacer refractory period of embodiments should not be confused with the refractory period exhibited by ECG pulses. Indeed, the heart is refrac-

tory to any stimulation following a QRS. For purposes of pacer embodiments, however, a pacer refractory period may occur immediately after a sensed beat or a pacing pulse. A pacer refractory period can be a “logical” period when any detected R-wave or P-wave is ignored. This should also not be confused with a blanking period; a blanking period is a time, nominally shorter than a pacer refractory period, where circuitry is disabled/blanked to minimize the interference from the actual pacing pulse.

[0133] In embodiments, the pacer refractory period of the QRS detector is coordinated with the heart’s refractory period. The T-wave of a driven complex may extend 400-500 msec after the pacing pulse. In some instances, therefore, the pacer refractory period, and time TM2, may need to be at least 500 msec after time TM1 to avoid detecting driven complexes.

[0134] Therefore, in some embodiments, analyzing the ECG signal may include detecting QRS complexes in the particular portion of the ECG signal during the active period P_ACTIVE, but not in the pacer refractory period P_PACER REFRACTORY. This way, in some embodiments, the continuance determination of the decision diamond 833 can optionally be made from the QRS complexes detected in the active period.

[0135] Returning again to FIG. 8, possible results of the decision diamond 833 are now described. In some embodiments, the decision can be YES. For instance, according to an optional internal decision diamond 840, it may be determined whether one or more criteria for change are met and, if not, then the decision YES of the diamond 833 can be a default one.

[0136] If the decision at the diamond 833 is to continue, the result is shown in the sample planned sequence 820-B, which is meant to be a continuation of the sample planned sequence 820-A. In particular, the planned sequence 820-B shows sample pacing pulses 826, 827, . . . against a time axis 829-B. These pacing pulses 826, 827, occur periodically, at the same regular intervals with spacings PACP1 as the pacing pulses in FIG. 820-A.

[0137] The continuing can be performed with no interruption to the planned sequence, or with a short interruption, e.g., where the time axis 829-A ends. For instance, the last planned pacing pulse 825 might not be caused to be delivered, and so on.

[0138] If the decision at the diamond 833 is to not continue, there can be a number of options, pointed to by multiple arrows.

[0139] One of the options is that pacing will continue, but the planned sequence will change—will be replaced with an updated sequence. In such instances, upon the discontinuing, the processor may start causing at least some of the electrical charge provided by the source to be discharged via at least two of the therapy electrodes through the ambulatory patient so as to continue delivering to the ambulatory patient an updated sequence of pacing pulses.

[0140] Such a result is shown in the sample changed or updated sequence 850, which is meant to continue after the planned sequence 820-A. In particular, the updated sequence 850 shows sample pacing pulses 856, 857, 858, . . . against a time axis 859. These pacing pulses 856, 857, 858 can be updated from those of the planned sequence in a number of ways.

[0141] In some embodiments, these pacing pulses 856, 857, 858, . . . are updated in that they have at least 10% more

energy than the pacing pulses of the planned sequence 820-A. In this example, the updated pulses of sequence 850 are shown taller, to indicate that they have more energy.

[0142] In some embodiments, the pacing pulses of the planned sequence 820-A have a first average period—here PACP1—and at least four of the pacing pulses in the updated sequence 850 (of which only three are shown) have a second average period at least 10% different than the first average period. In this example, the updated pulses occur faster, with a shorter sample period PACP2. In other instances, the more successful the pacing is, the pacing period may get longer—in other words slow down the pacing to allow more and more intrinsic heartbeats to be detected.

[0143] The updated sequence 850 may be implemented if an inadequacy criterion is used, with which to determine the adequacy or not of the planned pacing sequence. For instance, the continuance determination may include determining whether or not an inadequacy criterion is met, and the discontinuing may be performed responsive to determining that the inadequacy criterion is met, and where the discontinuing would be followed by the updated pacing 850. The inadequacy criterion may include that intrinsic heartbeats do not occur, and/or driven heartbeats do not occur, and so on.

[0144] Again, the continuing with the updated sequence 850 can be performed with no interruption to the planned sequence, or with a short interruption, e.g., where the time axis 829-A ends. For instance, the last planned pacing pulse 825 might not be caused to be delivered, and so on.

[0145] Another one of the options is to stop pacing altogether. In such embodiments, the discontinuing may include causing no more of the pacing pulses to be delivered for at least two minutes. Such a result is shown in the sample box 870. While a time axis 879 is shown, no pacing pulses are caused to be delivered.

[0146] This option of stopping pacing altogether may be implemented in a number of ways. In some embodiments, the continuance determination of the decision diamond 833 includes determining whether or not an escape criterion is met, and the discontinuing is performed responsive to determining that the escape criterion is met. In some embodiments, the processor is further configured to detect QRS complexes in the particular portion 812 of the ECG signal, and to compute a heart rate from the detected QRS complexes. In such embodiments, the escape criterion may include that the heart rate is larger than 30 beats per minute (bpm). In some of those embodiments, the QRS complexes have been detected during only the active period but not during the pacer refractory period of FIG. 10. In such embodiments, one of the pacing pulses in the planned sequence may be prevented from being delivered, responsive to detecting an intrinsic QRS complex. This way, if intrinsic QRS complexes take over, pacing may stop altogether. This is possible in embodiments thanks to the pauses. Otherwise, with fast pacing, the heart would have to beat even faster for escaping pacing. This would probably not be good for a WMS patient because a WMS pacing is desirable mostly if the patient’s rate is too low to sustain life.

[0147] One more option is that the WMS may defibrillate per the operation 880, in embodiments where a defibrillator is also provided. The operation 880 may be performed, for instance, from the diamond 833 if a defibrillation criterion is met at the optional internal diamond 840. This may be in

addition to defibrillating again later anyway. Other events, detections, etc. may or may not intervene.

[0148] FIG. 11 shows a flowchart 1100 for describing methods according to embodiments. The method can start at an operation 1110, where an ECG signal is sensed. This can be performed as described above.

[0149] According to another operation 1120, a planned sequence of pacing pulses can be caused to be delivered. In particular, at least some of the electrical charge provided by the source can be caused to be discharged via at least two of the therapy electrodes through the ambulatory patient so as to deliver to the ambulatory patient a planned sequence of pacing pulses. The planned sequence can be as described above. Plus, it may have at least 5 successive pacing pulses with four spacings between successive ones of them, no two of the at least 5 successive pacing pulses having a spacing between them that is longer than 4 times an average of the four spacings. Or 3 times, or 2 times, or 1.7 times or 1.4 times, as per the above. Optionally, and similarly to what was described previously, the ambulatory patient may be warned of the impending pacing prior to the operation 1120. In particular, a speaker of the UI 280 may output a warning prompt prior to the planned sequence of the pacing pulses of the operation 1120 being caused to be delivered. The processor can be further configured to determine whether or not the cancel switch has been actuated after the warning prompt has been output. In such embodiments, the planned sequence of the pacing pulses is not caused to be delivered responsive to determining that the cancel switch has been actuated after the warning prompt has been output.

[0150] According to another operation 1130, a particular portion of the ECG signal, which has been sensed after the delivery per the planned sequence starts and before it stops, may be analyzed. A conventional rhythm analysis with a very high sensitivity can be run, precluding the possibility that the patient may have undetected VF. In embodiments, simultaneous, multi-channel, real-time R-wave detection across all ECG channels that meet criteria for minimal noise and continuous electrode contact may be used. Once the detector refractory period is satisfied, an R-wave detection on any qualified ECG channel is considered an intrinsic beat for the purpose of inhibition. Moreover, the QRS detection may be turned off and on for each ECG channel's R-wave detector. Additionally, sensing refractory periods can be configurable, and thus allow refractory periods for intrinsic pulses to be different from refractory periods following pacing pulse delivery. Also, if a high number of paced heartbeats are occurring, consider the patient's rhythm to be bradycardic/asystolic since few or no intrinsic R-waves are being detected by the sensitive, multi-channel R-wave detectors during the long periods between pacing pulses (e.g., 1.5 sec at 40 bpm). If there are long periods of time without paced heartbeats (e.g., more than 3 sec implying at least 2 intrinsic R-waves have been detected), perform rhythm analysis on these pace-pulse-free ECG data sections.

[0151] According to another operation 1133, a continuance determination may be made from the analyzed particular portion of the ECG signal. Similarly with what was written for the decision diamond 833, the continuance determination may be as to whether or not to continue pacing according to the planned sequence.

[0152] If, at the operation 1133, the answer is yes, then causing at least some of the electrical charge provided by the source to be discharged via at least two of the therapy

electrodes through the ambulatory patient may be continued, so as to deliver to the ambulatory patient the planned sequence of pacing pulses for at least 10 more pacing pulses. That would be by the execution returning to operation 1110, and then 1120 for continuing the planned sequence.

[0153] If, at the operation 1133, the answer is no, then according to another operation 1134 the causing to be delivered to the ambulatory patient the planned sequence of pacing pulses may be discontinued.

[0154] If at the operation 1133 an inadequacy criterion 1135 is met then, according to another, optional operation 1150, pacing may be continued, but with different energy and/or with different average period, similarly with what was described above for the updated sequence 850.

[0155] A method of external pacing is to give a pacing pulse after refractory period of intrinsic QRS complex and then wait an interval equal to the intrinsic rate. If an intrinsic QRS complex is detected about the same intrinsic rate from the previous QRS complex, the pacing is not captured and increase the energy level then repeat the same previous procedure until the pacing is captured.

[0156] When a pacing pulse is captured, the interval from the pacing pulse to the following intrinsic QRS complex should be significantly longer than the previous pacing pulse which failed to capture to the intrinsic QRS complex.

[0157] Once the pacing threshold becomes known, the pacing level can be set at twice the pacing threshold and the above-mentioned methods can be applied.

[0158] The operation 1134 includes the optional operation 1170, where pacing may stop altogether for at least two minutes. That may be where an escape criterion 1137 is met, and so on, per the above.

[0159] If at the operation 1133 a defibrillation criterion 1138 is met then, according to another operation 1180, a defibrillation shock may be caused to be delivered, in embodiments where a defibrillator is also provided.

[0160] Another method of external pacing would be to apply pacing pulses and examine the ECG signal to detect the presence of artifacts. If the ECG is determined to be artifact-free, then intrinsic beats (QRS complexes) can be detected between pacing pulses. If excessive artifacts are detected, then pauses can be applied as described above to allow artifact-free analysis. In this scenario it is important that the QRS detector have a pacer refractory period after the pacing pulse to prevent detection of the pacing pulse or the driven complex. This method may work best with a relatively low pacing pulse rate, perhaps 60 bpm. At 60 bpm, a 500 msec pacer refractory period can be applied and the remaining 500 msec could be used for sensing intrinsic complexes.

[0161] On the other hand, if the pacing rate is set to 90 bpm, then the interval between pacing pulses is only 666 mS. If 500 msec is used for a pacer refractory period, then only 166 msec are available for sensing intrinsic complexes. That means that intrinsic complexes can be detected only about 25% of the time, which is not good if the pacing pulses have not achieved capture. If the patient's rate is the same (or similar) to the pacing rate, it is possible that all of the intrinsic beats could fall in the pacer refractory period and the intrinsic rate is never detected.

[0162] In some embodiments the refractory period could be different for different patients, and the pacer refractory period may be set accordingly. It is possible that a physician could set the pacer refractory period when the device is

prescribed, or perhaps the device could sense the end of the driven complex and start looking for intrinsic complexes at the end of the driven T-wave.

[0163] In the methods described above, each operation can be performed as an affirmative act or operation of doing, or causing to happen, what is written that can take place. Such doing or causing to happen can be by the whole system or device, or just one or more components of it. It will be recognized that the methods and the operations may be implemented in a number of ways, including using systems, devices and implementations described above. In addition, the order of operations is not constrained to what is shown, and different orders may be possible according to different embodiments. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Moreover, in certain embodiments, new operations may be added, or individual operations may be modified or deleted. The added operations can be, for example, from what is mentioned while primarily describing a different system, apparatus, device or method.

[0164] A person skilled in the art will be able to practice the present invention in view of this description, which is to be taken as a whole. Details have been included to provide a thorough understanding. In other instances, well-known aspects have not been described, in order to not obscure unnecessarily this description.

[0165] Some technologies or techniques described in this document may be known. Even then, however, it does not necessarily follow that it is known to apply such technologies or techniques as described in this document, or for the purposes described in this document.

[0166] This description includes one or more examples, but this fact does not limit how the invention may be practiced. Indeed, examples, instances, versions or embodiments of the invention may be practiced according to what is described, or yet differently, and also in conjunction with other present or future technologies. Other such embodiments include combinations and sub-combinations of features described herein, including for example, embodiments that are equivalent to the following: providing or applying a feature in a different order than in a described embodiment; extracting an individual feature from one embodiment and inserting such feature into another embodiment; removing one or more features from an embodiment; or both removing a feature from an embodiment and adding a feature extracted from another embodiment, while providing the features incorporated in such combinations and sub-combinations.

[0167] In general, the present disclosure reflects preferred embodiments of the invention. The attentive reader will note, however, that some aspects of the disclosed embodiments extend beyond the scope of the claims. To the respect that the disclosed embodiments indeed extend beyond the scope of the claims, the disclosed embodiments are to be considered supplementary background information and do not constitute definitions of the claimed invention.

[0168] In this document, the phrases “constructed to”, “adapted to” and/or “configured to” denote one or more actual states of construction, adaptation and/or configuration that is fundamentally tied to physical characteristics of the element or feature preceding these phrases and, as such, reach well beyond merely describing an intended use. Any such elements or features can be implemented in a number

of ways, as will be apparent to a person skilled in the art after reviewing the present disclosure, beyond any examples shown in this document.

[0169] Incorporation by reference: References and citations to other documents, such as patents, patent applications, patent publications, journals, books, papers, web contents, have been made throughout this disclosure. All such documents are hereby incorporated herein by reference in their entirety for all purposes.

[0170] Parent patent applications: Any and all parent, grandparent, great-grandparent, etc. patent applications, whether mentioned in this document or in an Application Data Sheet (“ADS”) of this patent application, are hereby incorporated by reference herein as originally disclosed, including any priority claims made in those applications and any material incorporated by reference, to the extent such subject matter is not inconsistent herewith.

[0171] Reference numerals: In this description a single reference numeral may be used consistently to denote a single item, aspect, component, or process. Moreover, a further effort may have been made in the preparation of this description to use similar though not identical reference numerals to denote other versions or embodiments of an item, aspect, component or process that are identical or at least similar or related. Where made, such a further effort was not required, but was nevertheless made gratuitously so as to accelerate comprehension by the reader. Even where made in this document, such a further effort might not have been made completely consistently for all of the versions or embodiments that are made possible by this description. Accordingly, the description controls in defining an item, aspect, component or process, rather than its reference numeral. Any similarity in reference numerals may be used to infer a similarity in the text, but not to confuse aspects where the text or other context indicates otherwise.

[0172] The claims of this document define certain combinations and subcombinations of elements, features and acts or operations, which are regarded as novel and non-obvious. The claims also include elements, features and acts or operations that are equivalent to what is explicitly mentioned. Additional claims for other such combinations and subcombinations may be presented in this or a related document. These claims are intended to encompass within their scope all changes and modifications that are within the true spirit and scope of the subject matter described herein. The terms used herein, including in the claims, are generally intended as “open” terms. For example, the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” etc. If a specific number is ascribed to a claim recitation, this number is a minimum but not a maximum unless stated otherwise. For example, where a claim recites “a” component or “an” item, it means that the claim can have one or more of this component or this item.

[0173] In construing the claims of this document, the inventor(s) invoke 35 U.S.C. § 112(f) only when the words “means for” or “steps for” are expressly used in the claims. Accordingly, if these words are not used in a claim, then that claim is not intended to be construed by the inventor(s) in accordance with 35 U.S.C. § 112(f).

What is claimed is:

1. A method of operating a wearable medical device (WMD) system worn by a patient, the method comprising:

sensing, by one or more ECG electrodes of the WMD system, an electrocardiogram (ECG) signal of the patient;

determining, based on the sensed ECG signal of the patient, whether the patient is experiencing a medical condition;

responsive to determining that the patient is experiencing the medical condition, causing at least some electrical charge stored in one or more energy sources of the WMD system to be discharged via at least two therapy electrodes through the patient, to deliver a sequence of pacing pulses to the patient;

analyzing, by a processor of the WMD system, a particular portion of the sensed ECG signal during the delivery of the sequence of pacing pulses;

determining whether to continue pacing according to the sequence of pacing pulses;

responsive to determining to continue pacing according to the sequence of pacing pulses, causing at least some of the electrical charge to be discharged via the at least two therapy electrodes through the patient to deliver the sequence of pacing pulses for at least a threshold period of time; and

responsive to determining to discontinue the pacing according to the sequence of pacing pulses:

causing a pause in the discharge of pacing pulses,

analyzing portions of the sensed ECG signal that were sensed during the pause against one or more escape criteria to determine whether the one or more escape criteria are met, and

responsive to determining that the one or more escape criteria are met, causing the one or more energy sources to discontinue the discharge of pacing pulses.

2. The method of claim 1, wherein determining whether to continue pacing according to the sequence of pacing pulses comprises:

determining whether a defibrillation criterion is met; and

responsive to determining that the defibrillation criterion is met, causing a defibrillation shock to be delivered to the patient.

3. The method of claim 1, wherein the sequence of pacing pulses comprises at least five successive pacing pulses having four spacings between successive ones of the pacing pulses, and wherein no two of the at least five successive pacing pulses have a spacing therebetween that is greater than four times an average of the four spacings.

4. The method of claim 1, further comprising:

prior to the delivery of the sequence of pacing pulses, outputting a warning prompt via a speaker of the WMD system; and

responsive to determining that a cancel switch has been actuated after the warning prompt has been output, aborting to deliver the sequence of pacing pulses.

5. The method of claim 1, wherein the analyzing the particular portion of the sensed ECG signal comprises R-wave detection across multiple ECG channels that meet criteria for minimal noise and continuous electrode contact.

6. The method of claim 5, wherein each of the multiple ECG channels includes a configurable refractory period, and a refractory period following delivery of a pacing pulse is different from a refractory period following detection of an intrinsic pulse.

7. The method of claim 1, wherein determining whether to continue pacing according to the sequence of pacing pulses comprises:

determining whether an inadequacy criterion is met; and

responsive to determining that the inadequacy criterion is met, continuing to deliver the sequence of pacing pulses by modifying at least one of an energy level or an average period of the pacing pulses.

8. A method of operating a wearable medical device (WMD) system worn by a patient, the method comprising:

sensing, by one or more ECG electrodes of the WMD system, an electrocardiogram (ECG) signal of the patient;

analyzing, by a processor of the WMD system, portions of the sensed ECG signal using bradycardia pacing criteria to determine whether the bradycardia pacing criteria are met;

causing, responsive at least in part to determining that the bradycardia pacing criteria are met, at least one of one or more energy sources to discharge at least some stored electrical charge as pacing pulses via one or more therapy electrodes while the patient is wearing the WMD system;

analyzing portions of the sensed ECG signal using asystole pacing criteria to determine whether the asystole pacing criteria are met;

causing, responsive at least in part to determining that the asystole pacing criteria are met, the at least one of the one or more energy sources to discharge at least some of the stored electrical charge as pacing pulses via the one or more therapy electrodes while the patient is wearing the WMD system;

causing a pause in the discharge of pacing pulses;

analyzing portions of the sensed ECG signal that were sensed during the pause against one or more escape pacing criteria to determine whether the one or more escape pacing criteria are met; and

responsive to determining that the one or more escape pacing criteria are met, causing the at least one of the one or more energy sources to discontinue the discharge of pacing pulses.

9. The method of claim 8, further comprising:

analyzing portions of the sensed ECG signal using shock criteria to determine whether the shock criteria are met; and

causing, responsive at least in part to determining that the shock criteria are met, at least one of the one or more energy sources to discharge at least some of the stored electrical charge as a shock via the one or more therapy electrodes while the patient is wearing the WMD system.

10. The method of claim 8, wherein the bradycardia pacing criteria comprise a first bradycardia criterion for an onset of a bradycardia event, and a second bradycardia criterion that comprises a confirmation period to confirm the bradycardia event, the confirmation period occurring after the first bradycardia criterion is met.

11. The method of claim 10, wherein at least some of the portions of the sensed ECG signal analyzed against the first bradycardia criterion are different from portions of the sensed ECG signal analyzed against the second bradycardia criterion.

12. The method of claim 10, wherein at least some of the portions of the sensed ECG signal analyzed against the second bradycardia criterion were sensed during the confirmation period.

13. The method of claim 10, wherein the first bradycardia criterion comprises a heart rate of less than 40 beats per minute (bpm).

14. The method of claim 10, wherein the confirmation period lasts at least 20 seconds.

15. The method of claim 10, wherein the second bradycardia criterion includes the first bradycardia criterion.

16. The method of claim 8, wherein the discontinuation of the discharge of pacing pulses lasts for at least sixty seconds.

17. The method of claim 8, wherein the one or more escape pacing criteria includes a heart rate of at least 30 beats per minute (bpm).

18. The method of claim 8, wherein the pacing pulses have a pacing rate between 50 beats per minute (bpm) and 90 bpm.

19. The method of claim 8, further comprising causing a speaker of the WMD system to output a warning prompt prior to causing the pacing pulses to be discharged.

20. The method of claim 19, further comprising:
detecting an input, at a user interface of the WMD system,
after the warning prompt has been output; and
responsive to detecting the input, causing the at least one
of the one or more energy sources to not discharge the
pacing pulses.

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